Modeling Approaches on the FtsZ-AmpR-AmpC axis

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Contents

1 Hypothesis **4**

2 Model Architecture **4**

2.1 Peptidoglycan Biosynthesis 4

2.1.1 UDP-NAG (Uridine diphosphate-N-acetylglucosamine) 5

2.1.2 UDP-NAG-enolpyruvate 5

2.1.3 UDP-NAM (Uridine diphosphate-N-acetylmuramic acid) 6

2.1.4 UDP-NAM-L-Ala 6

2.1.5 UDP-NAM-L-Ala-γ-D-Glu 7

2.1.6 UDP-NAM-L-Ala-γ-D-Glu-m-DAP 7

2.1.7 UDP-NAM-pentapeptide 8

2.1.8 Lipid I (Cytoplasmic face) 9

2.1.9 Lipid II (Cytoplasmic face) 9

2.2 Periplasmic Reactions 9

2.2.1 Lipid II (Periplasmic face) 10

2.2.2 Uncrosslinked Pentapeptide Peptidoglycan (PG_{UP}) . 12

2.2.3 Uncrosslinked Tetrapeptide Peptidoglycan (PG_{UT}) . . 14

2.2.4 Crosslinked Peptidoglycan (PG_{XL}) 14

2.2.5 Periplasmic Pentapeptide Disaccharide ($Disacch_{P,peri}$) - NAG-1,6-anh-NAM-pentapeptide 15

2.2.6 Periplasmic Tetrapeptide Disaccharide ($Disacch_{T,peri}$) - NAG-1,6-anh-NAM-tetrapeptide 15

2.3 Peptidoglycan Recycling (Cytoplasmic Reactions) 16

2.3.1 Cytoplasmic Pentapeptide Disaccharide ($Disacch_{P,cyto}$) - NAG-1,6-anh-NAM-pentapeptide 16

2.3.2 Cytoplasmic Tetrapeptide Disaccharides ($Disacch_{T,cyto}$) - NAG-1,6-anh-NAM-tetrapeptide . . . 17
2.3.3 Cytoplasmic Tripeptide Disaccharides ($Disacch_{Tri}$) - NAG-1,6-anh-NAM-tripeptide . . . 17
2.3.4 Peptide-free Disaccharides ($Disacch$) - NAG-1,6-anh-NAM . . . 18
2.3.5 1,6-anh-NAM-pentapeptide (Anh_P) - AmpR Activating Muropeptide . . . 18
2.3.6 1,6-anh-NAM-tetrapeptide (Anh_T) . . . 19
2.3.7 1,6-anh-NAM-tripeptide (Anh_{Tri}) . . . 19
2.3.8 1,6-anh-NAM ($anhNAM$) . . . 20
2.3.9 NAM-6-Phosphate (NAM-6-P) . . . 20
2.3.10 NAG-6-Phosphate ($NAG-6-P$) . . . 21
2.3.11 Glucosamine 6-phosphate ($GlcN-6-P$) . . . 21
2.3.12 Glucosamine 1-phosphate ($GlcN-1-P$) . . . 22
2.3.13 N-Acetyl Glucosamine (NAG) . . . 22
2.3.14 Free Pentapeptide Stem ($Stem_{Penta}$) . . . 23
2.3.15 Free Tetrapeptide Stem ($Stem_{Tetra}$) . . . 23
2.3.16 Free Tripeptide Stem ($Stem_{Tri}$) . . . 24
2.3.17 Und-P Recycling Reactions . . . 25
2.3.18 Periplasmic Undecaprenyl Pyrophosphate ($Und-PP_{peri}$) 25
2.3.19 Periplasmic Undecaprenyl Phosphate ($Und-P_{peri}$) . 26
2.3.20 Cytoplasmic Undecaprenyl Phosphate ($Und-P_{cyto}$) . 26
2.4 AmpR Regulation and AmpC Production . . . 27
2.4.1 AmpR transcription and translation . . . 27
2.4.2 AmpC transcription and translation . . . 28
2.5 β-Lactam and PBP Kinetics . . . 29
2.5.1 β-Lactam . . . 29
2.5.2 PBPs . . . 30
2.6 FtsZ Kinetics . . . 30

3 Parameters **31**

4 Test Conditions **31**
4.1 The Four Experimental Conditions . . . 32

5 Simulation Methodology **33**

6 Parameter Estimation **33**
6.1 Manual Calibration 34
6.2 Automated Calibration 34

7 Testing the Calibrated Model on Different Drugs **35**

8 Results **35**
8.1 Simulation Results of Original, Uncalibrated Model 35
8.2 Simulation Results of Manually Calibrated Model 37
8.2.1 Parameters Calibrated 37
8.2.2 Results after Calibration 38
8.2.3 AmpC Fold-Induction: Model versus Literature 39
8.3 Simulation Results of Automated Calibrated Model 40
8.4 Simulation Results of the Model Tested on Different Drugs . 42

1 Hypothesis

Depletion or inhibition of a critical cell division protein, FtsZ, restores sensitivity to β-lactams in Gram-Negative bacteria possessing an inducible β-lactamase (AmpC) operon. AmpR is the DNA-binding regulatory protein that governs the expression of AmpC by binding to different muropeptides and modulating the rate of transcription of AmpC. Under exposure of β-lactams, there is an increased turnover of activator muropeptides recycled from the affected cell wall that transform the AmpR to an activator, thereby inducing AmpC expression and neutralising the β-lactam threat. However, in FtsZ mutants, inactivation of FtsZ leads to an accumulation of UDP-NAM-pentapeptide, the repressor muropeptide, that binds to AmpR, converts it to its repressor state, and thereby switches off AmpC expression even in the presence of β-lactams.

This has been observed first by Ottolenghi and Ayala (1991), with further validation and identification of the repressor muropeptide done by Uehara and Park (2002). This study aims to recreate the results of these experiments in silico through mathematical modelling of the involved pathways, and serve as a motivation to experiment along these lines on native *Pseudomonas aeruginosa*.

2 Model Architecture

We are modelling an *E. coli* system which has been transformed with a plasmid bearing *C. freundii* AmpR-AmpC genes. This is slightly different from the previous model we built on *P. aeruginosa* by a few additional reactions in the murein recycling pathway. FtsZ function is controlled through a tunable parameter Z.

The FtsZ-AmpR-AmpC axis encompasses a network of three pathways - **peptidoglycan biosynthesis**, **peptidoglycan recycling** and **β-lactam resistance (AmpC induction)**. All reactions are grouped and depicted in the following sections.

2.1 Peptidoglycan Biosynthesis

The following equations represent a mathematical model of the Mur pathway, the sequential cytoplasmic enzymatic cascade that bacteria use to synthesize Lipid II, the essential building block of the peptidoglycan cell wall.

Every equation follows a standard mass-balance structure: the rate of change in a metabolite's concentration $\frac{d[X]}{dt}$ is equal to its rate of synthesis

minus its rate of consumption and its rate of dilution due to bacterial cell volume growth (represented in every equation by the term: μ[Metabolite].

2.1.1 UDP-NAG (Uridine diphosphate-N-acetylglucosamine)

$$J_{GlmU} = \frac{V_{max,GlmU} \cdot [GlcN-1-P]}{K_{m,GlmU} + [GlcN-1-P]} \quad \text{(i)}$$

$$J_{MurA} = \frac{V_{max,MurA} \cdot [UDP\text{-}NAG]}{K_{m,MurA} + [UDP\text{-}NAG]} \cdot \frac{1}{1 + \left(\frac{[UDP\text{-}NAM]}{K_{inhibit}}\right)^n} \quad \text{(ii)}$$

$$J_{MurG} = \frac{V_{max,MurG} \cdot [\text{Lipid I}_{cyto}] \cdot [UDP\text{-}NAG]}{(K_{m,LipidI} + [\text{Lipid I}_{cyto}])(K_{m,UDP-NAG} + [UDP\text{-}NAG])} \quad \text{(iii)}$$

$$\frac{d[UDP\text{-}NAG]}{dt} = v_{in,NAG} + J_{GlmU} - J_{MurA} - J_{MurG} - \mu[UDP\text{-}NAG] \quad (1)$$

This equation tracks the primary precursor, **UDP-NAG**, the first metabolite of the pathway. It is continuously supplied to the system at a basal rate ($v_{in,NAG}$) and by the peptidoglycan recycling pathway (J_{GlmU}). It is consumed in two different places in the pathway:

- By **MurA** at the start of the pathway. Notably, this reaction features a Hill-equation multiplier representing **feedback inhibition**. If downstream UDP-NAM builds up, it inhibits MurA to prevent overproduction [Mizyed et al. (2005)].
- By **MurG** near the end of the pathway, where UDP-NAG is combined with Lipid I. This is modeled using bi-substrate Michaelis-Menten kinetics.

2.1.2 UDP-NAG-enolpyruvate

$$J_{MurA} = \frac{V_{max,MurA} \cdot [UDP\text{-}NAG]}{K_{m,MurA} + [UDP\text{-}NAG]} \cdot \frac{1}{1 + \left(\frac{[UDP\text{-}NAM]}{K_{inhibit}}\right)^n} \quad \text{(i)}$$

$$J_{MurB} = \frac{V_{max,MurB} \cdot [UDP\text{-}NAG\text{-}enolpyruvate]}{K_{m,MurB} + [UDP\text{-}NAG\text{-}enolpyruvate]} \quad \text{(ii)}$$

$$\frac{d[UDP\text{-}NAG\text{-}enolpyruvate]}{dt} = J_{MurA} - J_{MurB} - \mu[UDP\text{-}NAG\text{-}enolpyruvate] \quad (2)$$

This tracks the intermediate produced by **MurA**. Its synthesis perfectly matches the MurA consumption term from the first equation. It is subsequently consumed by the enzyme **MurB**, which reduces it to form UDP-NAM using standard single-substrate Michaelis-Menten kinetics.

2.1.3 UDP-NAM (Uridine diphosphate-N-acetylmuramic acid)

$$J_{MurB} = \frac{V_{max,MurB} \cdot [UDP\text{-}NAG\text{-}enolpyruvate]}{K_{m,MurB} + [UDP\text{-}NAG\text{-}enolpyruvate]} \tag{i}$$

$$J_{MurC} = \frac{V_{max,MurC} \cdot [UDP\text{-}NAM]}{K_{m,MurC} + [UDP\text{-}NAM]} \tag{ii}$$

$$J_{Mpl,Penta} = V_{max,Mpl,Penta} \cdot \left(\frac{[UDP\text{-}NAM]}{K_{M,Mpl,NAM} + [UDP\text{-}NAM]}\right) \cdot \left(\frac{[Stem_{Penta}]}{K_{M,Mpl,Penta} + [Stem_{Penta}]}\right) \tag{iii}$$

$$J_{Mpl,Tri} = V_{max,Mpl,Tri} \cdot \left(\frac{[UDP\text{-}NAM]}{K_{M,Mpl,NAM} + [UDP\text{-}NAM]}\right) \cdot \left(\frac{[Stem_{Tri}]}{K_{M,Mpl,Tri} + [Stem_{Tri}]}\right) \tag{iv}$$

$$\frac{d[UDP\text{-}NAM]}{dt} = J_{MurB} - J_{MurC} - J_{Mpl,Penta} - J_{Mpl,Tri} - \mu[UDP\text{-}NAM] \tag{3}$$

This mass-balance equation tracks the concentration of the central peptidoglycan precursor UDP-NAM. It is synthesized *de novo* by **MurB** (J_{MurB}) and consumed via two distinct pathways: the stepwise de novo assembly initiated by **MurC** (J_{MurC}), and the cell wall recycling pathway, where **Mpl** directly ligates salvaged intact pentapeptides ($J_{Mpl,Penta}$) or tripeptides ($J_{Mpl,Tri}$) using bisubstrate Michaelis-Menten kinetics.

2.1.4 UDP-NAM-L-Ala

$$J_{MurC} = \frac{V_{max,MurC} \cdot [UDP\text{-}NAM]}{K_{m,MurC} + [UDP\text{-}NAM]} \tag{i}$$

$$J_{MurD} = \frac{V_{max,MurD} \cdot [UDP\text{-}NAM\text{-}L\text{-}Ala]}{K_{m,MurD} + [UDP\text{-}NAM\text{-}L\text{-}Ala]} \tag{ii}$$

$$\frac{d[UDP\text{-}NAM\text{-}L\text{-}Ala]}{dt} = J_{MurC} - J_{MurD} - \mu[UDP\text{-}NAM\text{-}L\text{-}Ala] \tag{4}$$

This equation models the addition of the first amino acid to the stem peptide. **MurC** creates this molecule by adding L-Alanine, and **MurD** consumes it to add the next amino acid. Both follow standard Michaelis-Menten kinetics.

2.1.5 UDP-NAM-L-Ala-γ-D-Glu

$$J_{MurD} = \frac{V_{max,MurD} \cdot [UDP\text{-}NAM\text{-}L\text{-}Ala]}{K_{m,MurD} + [UDP\text{-}NAM\text{-}L\text{-}Ala]} \quad \text{(i)}$$

$$J_{MurE} = \frac{V_{max,MurE} \cdot [UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu]}{K_{m,MurE} + [UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu]} \quad \text{(ii)}$$

$$\frac{d[UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu]}{dt} = J_{MurD} - J_{MurE} - \mu[UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu] \quad (5)$$

Here, the dipeptide intermediate is formed by **MurD** (which adds D-Glutamate) and consumed by **MurE**.

2.1.6 UDP-NAM-L-Ala-γ-D-Glu-m-DAP

$$J_{MurE} = \frac{V_{max,MurE} \cdot [UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu]}{K_{m,MurE} + [UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu]} \quad \text{(i)}$$

$$J_{Mpl,Tri} = V_{max,Mpl,Tri} \cdot \left(\frac{[UDP\text{-}NAM]}{K_{M,Mpl,NAM} + [UDP\text{-}NAM]}\right) \cdot \left(\frac{[Stem_{Tri}]}{K_{M,Mpl,Tri} + [Stem_{Tri}]}\right) \quad \text{(ii)}$$

$$J_{MurF} = \frac{V_{max,MurF} \cdot [UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu\text{-}m\text{-}DAP]}{K_{m,MurF} + [UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu\text{-}m\text{-}DAP] + \frac{[UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu\text{-}m\text{-}DAP]^2}{K_{i,MurF}}} \quad \text{(iii)}$$

$$\frac{d[UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu\text{-}m\text{-}DAP]}{dt} = J_{MurE} + J_{Mpl,Tri} - J_{MurF} - \mu[UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu\text{-}m\text{-}DAP] \quad (6)$$

This equation tracks the tripeptide intermediate UDP-NAM-L-Ala-γ-D-Glu-m-DAP. It is generated through two converging pathways: the *de novo* addition of m-DAP by **MurE** (J_{MurE}) and the direct recycling of

salvaged tripeptides by **Mpl** ($J_{Mpl,Tri}$). It is consumed by **MurF** (J_{MurF}), which ligates a D-Ala-D-Ala dipeptide to finish the stem. The J_{MurF} flux is modeled using Haldane substrate inhibition kinetics (via the squared term and $K_{i,MurF}$), capturing the biological reality that excessive accumulation of this substrate impedes the enzyme's efficiency [Anderson et al. (1996)].

2.1.7 UDP-NAM-pentapeptide

$$J_{MurF} = \frac{V_{max,MurF} \cdot [UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu\text{-}m\text{-}DAP]}{K_{m,MurF} + [UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu\text{-}m\text{-}DAP] + \frac{[UDP\text{-}NAM\text{-}L\text{-}Ala\text{-}\gamma\text{-}D\text{-}Glu\text{-}m\text{-}DAP]^2}{K_{i,MurF}}} \tag{i}$$

$$J_{Mpl,Penta} = V_{max,Mpl,Penta} \cdot \left(\frac{[UDP\text{-}NAM]}{K_{M,Mpl,NAM} + [UDP\text{-}NAM]}\right) \cdot \left(\frac{[Stem_{Penta}]}{K_{M,Mpl,Penta} + [Stem_{Penta}]}\right) \tag{ii}$$

$$J_{MraY} = V_{max,MraY} \cdot \left(\frac{[UDP\text{-}NAM\text{-}pentapeptide]}{K_{M,UDP\text{-}NAM\text{-}pentapeptide} + [UDP\text{-}NAM\text{-}pentapeptide]}\right) \cdot \left(\frac{[\text{Und-P}_{cyto}]}{K_{M,UndP} + [\text{Und-P}_{cyto}]}\right) \tag{iii}$$

$$\frac{d[UDP\text{-}NAM\text{-}pentapeptide]}{dt} = J_{MurF} + J_{Mpl,Penta} - J_{MraY} - \mu[UDP\text{-}NAM\text{-}pentapeptide] \tag{7}$$

This equation tracks UDP-NAM-pentapeptide, the final soluble cytosolic precursor (and the key repressor of AmpC). It accumulates via the *de novo* ligation of D-Ala-D-Ala by **MurF** (J_{MurF}) and the recycling of intact pentapeptides by **Mpl** ($J_{Mpl,Penta}$). It is consumed by **MraY** (J_{MraY}), an integral membrane enzyme that anchors it to the lipid carrier undecaprenyl phosphate (Und-P_{cyto}) to form Lipid I, modeled here with bisubstrate Michaelis-Menten kinetics.

2.1.8 Lipid I (Cytoplasmic face)

$$J_{MraY} = V_{max,MraY} \cdot \left(\frac{[UDP\text{-}NAM\text{-}pentapeptide]}{K_{M,UDP\text{-}NAM\text{-}pentapeptide} + [UDP\text{-}NAM\text{-}pentapeptide]} \right) \cdot \left(\frac{[\text{Und-P}_{cyto}]}{K_{M,UndP} + [\text{Und-P}_{cyto}]} \right) \quad \text{(i)}$$

$$J_{MurG} = \frac{V_{max,MurG} \cdot [\text{Lipid I}_{cyto}] \cdot [UDP\text{-}NAG]}{(K_{m,LipidI} + [\text{Lipid I}_{cyto}])(K_{m,UDP\text{-}NAG} + [UDP\text{-}NAG])} \quad \text{(ii)}$$

$$\frac{d[\text{Lipid I}_{cyto}]}{dt} = J_{MraY} - J_{MurG} - \mu[\text{Lipid I}_{cyto}] \quad (8)$$

MraY's successful attachment of the pentapeptide to the lipid carrier creates **Lipid I**. Next, **MurG** consumes Lipid I by linking it to the UDP-NAG (which we tracked in Equation 1). This is again a bi-substrate reaction.

2.1.9 Lipid II (Cytoplasmic face)

$$J_{MurG} = \frac{V_{max,MurG} \cdot [\text{Lipid I}_{cyto}] \cdot [UDP\text{-}NAG]}{(K_{m,LipidI} + [\text{Lipid I}_{cyto}])(K_{m,UDP\text{-}NAG} + [UDP\text{-}NAG])} \quad \text{(i)}$$

$$J_{MurJ} = \frac{V_{max,MurJ} \cdot [\text{Lipid II}_{cyto}]}{K_{M,MurJ} + [\text{Lipid II}_{cyto}]} \quad \text{(ii)}$$

$$\frac{d[\text{Lipid II}_{cyto}]}{dt} = J_{MurG} - J_{MurJ} - \mu[\text{Lipid II}_{cyto}] \quad (9)$$

MurG synthesizes **Lipid II**, the final complete monomer for the cell wall. Once assembled on the inner leaf of the cytoplasmic membrane, it is consumed by the flippase enzyme **MurJ**, which transports Lipid II across the membrane to the outside of the cell so it can be polymerized into the actual peptidoglycan wall.

2.2 Periplasmic Reactions

These differential equations model the periplasmic (extracellular space between the inner and outer membranes) stages of the peptidoglycan cell wall construction and breakdown. The models account for polymerization, cross-linking, cell division regulation, and degradation.

2.2.1 Lipid II (Periplasmic face)

$$J_{MurJ} = \left(\frac{V_{cyto}}{V_{peri}}\right) \frac{V_{max,MurJ}[\text{Lipid II}_{cyto}]}{K_{M,MurJ} + [\text{Lipid II}_{cyto}]} \tag{i}$$

$$J_{RodA} = \left(\frac{[\text{PBP2}]}{\text{PBP2}_{ss}}\right) \frac{V_{max,RodA}[\text{Lipid II}_{peri}]}{K_{M,RodA} + [\text{Lipid II}_{peri}]} \tag{ii}$$

$$J_{FtsW} = Z\left(\frac{[\text{PBP3}]}{\text{PBP3}_{ss}}\right) \frac{V_{max,FtsW}[\text{Lipid II}_{peri}]}{K_{M,FtsW} + [\text{Lipid II}_{peri}]} \tag{iii}$$

$$J_{PBP1a} = \frac{k_{cat,PBP1a,GT}[\text{PBP1a}][\text{Lipid II}_{peri}]}{K_{M,PBP1a,GT} + [\text{Lipid II}_{peri}]} \tag{iv}$$

$$J_{PBP1b} = Z\frac{k_{cat,PBP1b,GT}[\text{PBP1b}][\text{Lipid II}_{peri}]}{K_{M,PBP1b,GT} + [\text{Lipid II}_{peri}]} \tag{v}$$

$$\begin{aligned}\frac{d[\text{Lipid II}_{peri}]}{dt} = {} & J_{MurJ} - J_{RodA} - J_{FtsW} - J_{PBP1a} - J_{PBP1b} \\ & - \mu[\text{Lipid II}_{peri}]\end{aligned} \tag{10}$$

This equation tracks Lipid II after it has been "flipped" from the cytoplasm to the periplasm by **MurJ**.

- **Volume Adjustment:** The first term includes $(\frac{V_{cyto}}{V_{peri}})$ because the volume of the periplasm is significantly smaller than the cytoplasm; flipping the same number of molecules results in a higher concentration jump in the periplasm.
- **Polymerization:** Lipid II is consumed (polymerized into long chains) by four distinct glycosyltransferases:
  - **RodA (Elongasome):** Drives cell elongation (growing longer). Because RodA only functions when physically bound to PBP2 [Rohs et al. (2018)], its activity is scaled by $\frac{[PBP2]}{PBP2_{ss}}$ the fraction of surviving, uninhibited PBP2. If beta-lactam antibiotics irreversibly jam PBP2, this fraction approaches zero, physically halting RodA's ability to consume Lipid II.
  - **FtsW (Divisome):** Drives cell division (building the septum to split the cell). Similarly, FtsW only works when coupled to PBP3 [Taguchi et al. (2019)], so it is scaled by the fraction of uninhibited PBP3 $\frac{[PBP3]}{PBP3_{ss}}$. The FtsW term is also multiplied

by a state variable Z, representing the active divisome complex (e.g., FtsZ ring formation). If $Z = 0$ (the cell is not actively dividing) or if a β-Lactam drug neutralizes PBP3, FtsW activity stops entirely.

- **PBP1a (Constitutive Flux):** PBP1a is a Class A bifunctional enzyme, but this equation captures only its glycosyltransferase (GT) activity. It continuously polymerizes periplasmic Lipid II into new glycan strands for standard cell elongation, operating independently of the division cycle.
- **PBP1b (Division-Dependent Flux):** Similarly, this term models only the GT activity of the bifunctional PBP1b. Unlike PBP1a, its polymerization flux is gated by the Z-ring state (Z), driving localized Lipid II consumption explicitly during active septation.

2.2.2 Uncrosslinked Pentapeptide Peptidoglycan (PG_{UP})

$$A = \frac{[\mathrm{PG}_{UP}] + [\mathrm{PG}_{UT}]}{K_{M,acc} + [\mathrm{PG}_{UP}] + [\mathrm{PG}_{UT}]} \tag{i}$$

$$J_{RodA} = \left(\frac{[\mathrm{PBP2}]}{\mathrm{PBP2}_{ss}}\right) \frac{V_{max,RodA}[\text{Lipid II}_{peri}]}{K_{M,RodA} + [\text{Lipid II}_{peri}]} \tag{ii}$$

$$J_{FtsW} = Z\left(\frac{[\mathrm{PBP3}]}{\mathrm{PBP3}_{ss}}\right) \frac{V_{max,FtsW}[\text{Lipid II}_{peri}]}{K_{M,FtsW} + [\text{Lipid II}_{peri}]} \tag{iii}$$

$$J_{PBP1a,GT} = \frac{k_{cat,PBP1a,GT}[\mathrm{PBP1a}][\text{Lipid II}_{peri}]}{K_{M,PBP1a,GT} + [\text{Lipid II}_{peri}]} \tag{iv}$$

$$J_{PBP1b,GT} = Z\frac{k_{cat,PBP1b,GT}[\mathrm{PBP1b}][\text{Lipid II}_{peri}]}{K_{M,PBP1b,GT} + [\text{Lipid II}_{peri}]} \tag{v}$$

$$\begin{aligned} J_{TP} = A\Bigg[&\frac{k_{cat,PBP2}[\mathrm{PBP2}][\mathrm{PG}_{UP}]}{K_{M,PBP2} + [\mathrm{PG}_{UP}]} + Z\frac{k_{cat,PBP3}[\mathrm{PBP3}][\mathrm{PG}_{UP}]}{K_{M,PBP3} + [\mathrm{PG}_{UP}]} \\ &+ \frac{k_{cat,PBP1a,TP}[\mathrm{PBP1a}][\mathrm{PG}_{UP}]}{K_{M,PBP1a,TP} + [\mathrm{PG}_{UP}]} + Z\frac{k_{cat,PBP1b,TP}[\mathrm{PBP1b}][\mathrm{PG}_{UP}]}{K_{M,PBP1b,TP} + [\mathrm{PG}_{UP}]}\Bigg] \end{aligned} \tag{vi}$$

$$J_{PBP5} = \frac{k_{cat,PBP5}[\mathrm{PBP5}][\mathrm{PG}_{UP}]}{K_{M,PBP5} + [\mathrm{PG}_{UP}]} \tag{vii}$$

$$J_{PBP6} = \frac{k_{cat,PBP6}[\mathrm{PBP6}][\mathrm{PG}_{UP}]}{K_{M,PBP6} + [\mathrm{PG}_{UP}]} \tag{viii}$$

$$J_{PBP4,CP} = \frac{k_{cat,PBP4,CP}[\mathrm{PBP4}][\mathrm{PG}_{UP}]}{K_{M,PBP4,CP} + [\mathrm{PG}_{UP}]} \tag{ix}$$

$$J_{Slt} = \frac{V_{max,Slt}[\mathrm{PG}_{UP}]}{K_{M,Slt} + [\mathrm{PG}_{UP}]} \tag{x}$$

$$J_{Ami,UP} = \frac{V_{max,Ami,UP}[\mathrm{PG}_{UP}]}{K_{M,Ami,UP} + [\mathrm{PG}_{UP}]} \tag{xi}$$

$$\begin{aligned} \frac{d[\mathrm{PG}_{UP}]}{dt} = &J_{RodA} + J_{FtsW} + J_{PBP1a,GT} + J_{PBP1b,GT} - J_{TP} \\ &- J_{PBP5} - J_{PBP6} - J_{PBP4,CP} - J_{Slt} - J_{Ami,UP} \\ &- \mu[\mathrm{PG}_{UP}] \end{aligned} \tag{11}$$

Once Lipid II monomers are polymerized into long glycan chains, the resulting polymer is uncrosslinked peptidoglycan (PG_{UP}) harboring pentapeptide stems. This intermediate is highly transient and must be rapidly

processed to form the mature, rigid mesh structure of the cell wall or be cleaved for remodeling.

- **Synthesis:** PG_{UP} is generated by the glycosyltransferase (GT) activity of four distinct enzymes (matching the consumption terms from the periplasmic Lipid II equation):
  - The SEDS proteins **RodA** (elongation) and **FtsW** (division, gated by Z).
  - The bifunctional Class A enzymes **PBP1a** (constitutive elongation) and **PBP1b** (division, gated by Z).
- **Cross-linking (Consumption):** PG_{UP} is consumed to form cross-links via the composite transpeptidation flux (J_{TP}). This flux incorporates the TP activity of **PBP2** (elongation), **PBP3** (division, gated by Z), and the bifunctional **PBP1a/PBP1b**. This entire composite rate is scaled by A, a dimensionless term representing the localized availability of adjacent acceptor stems (PG_{UP} and PG_{UT}).
- **Carboxypeptidation:** To restrict further cross-linking, the terminal D-alanine of the pentapeptide stem is cleaved by DD-carboxypeptidases, specifically **PBP5, PBP6, and PBP4**, which convert PG_{UP} into tetrapeptide peptidoglycan (PG_{UT}).
- **Degradation:** A portion of PG_{UP} is naturally degraded to facilitate cell wall remodeling and daughter cell separation. This is driven by lytic transglycosylases (**Slt**) which cleave the glycan backbone, and amidases (**Ami**) which detach the peptide stems.

2.2.3 Uncrosslinked Tetrapeptide Peptidoglycan (PG_{UT})

$$J_{PBP4,EP} = \frac{k_{cat,PBP4,EP}[\text{PBP4}][\text{PG}_{XL}]}{K_{M,PBP4,EP} + [\text{PG}_{XL}]} \quad \text{(i)}$$

$$J_{PBP7} = \frac{k_{cat,PBP7}[\text{PBP7}][\text{PG}_{XL}]}{K_{M,PBP7} + [\text{PG}_{XL}]} \quad \text{(ii)}$$

$$J_{endo} = J_{PBP4,EP} + J_{PBP7} \quad \text{(iii)}$$

$$J_{PBP5} = \frac{k_{cat,PBP5}[\text{PBP5}][\text{PG}_{UP}]}{K_{M,PBP5} + [\text{PG}_{UP}]} \quad \text{(iv)}$$

$$J_{PBP6} = \frac{k_{cat,PBP6}[\text{PBP6}][\text{PG}_{UP}]}{K_{M,PBP6} + [\text{PG}_{UP}]} \quad \text{(v)}$$

$$J_{PBP4,CP} = \frac{k_{cat,PBP4,CP}[\text{PBP4}][\text{PG}_{UP}]}{K_{M,PBP4,CP} + [\text{PG}_{UP}]} \quad \text{(vi)}$$

$$J_{Slt} = \frac{V_{max,Slt}[\text{PG}_{UT}]}{K_{M,Slt} + [\text{PG}_{UT}]} \quad \text{(vii)}$$

$$J_{Ami,UT} = \frac{V_{max,Ami,UT}[\text{PG}_{UT}]}{K_{M,Ami,UT} + [\text{PG}_{UT}]} \quad \text{(viii)}$$

$$\begin{aligned}\frac{d[\text{PG}_{UT}]}{dt} = {} & J_{PBP5} + J_{PBP6} + J_{PBP4,CP} + J_{endo} - J_{Slt} - J_{Ami,UT} \\ & - \mu[\text{PG}_{UT}]\end{aligned} \quad (12)$$

This equation isolates the uncrosslinked peptidoglycan network harboring tetrapeptide stems (PG_{UT}). It is synthesized through two distinct pathways: the carboxypeptidase activity of **PBP5, PBP6, and PBP4** (which trim the terminal D-alanine from PG_{UP} stems), and the endopeptidase activity (J_{endo}) of **PBP4 and PBP7** (which cleave mature, crosslinked peptidoglycan, PG_{XL}). It tracks the continuous turnover and degradation of this intermediate by lytic transglycosylases (J_{Slt}) and amidases ($J_{Ami,UT}$).

2.2.4 Crosslinked Peptidoglycan (PG_{XL})

$$J_{Ami,XL} = \frac{V_{max,Ami,XL}[\text{PG}_{XL}]}{K_{M,Ami,XL} + [\text{PG}_{XL}]} \quad \text{(i)}$$

$$\begin{aligned}\frac{d[\text{PG}_{XL}]}{dt} = {} & J_{TP} - J_{endo} - J_{Ami,XL} \\ & - \mu[\text{PG}_{XL}]\end{aligned} \quad (13)$$

This equation tracks mature, crosslinked peptidoglycan (PG_{XL}), the primary polymer that provides structural integrity to the bacterial cell wall. It is synthesized via the transpeptidation cross-linking flux (J_{TP}) acting on adjacent uncrosslinked stems. Because this rigid mesh must flexibly expand to accommodate cell growth and division, it is continuously remodeled: endopeptidases (J_{endo}) cleave the peptide cross-links (reverting it to PG_{UT}), while amidases ($J_{Ami,XL}$) detach the crosslinked peptide stems directly from the glycan backbone.

2.2.5 Periplasmic Pentapeptide Disaccharide ($Disacch_{P,peri}$) - NAG-1,6-anh-NAM-pentapeptide

$$J_{Slt} = \frac{V_{max,Slt}[\text{PG}_{UP}]}{K_{M,Slt} + [\text{PG}_{UP}]} \tag{i}$$

$$J_{AmpG} = \frac{V_{max,AmpG}[\text{Disacch}_{P,peri}]}{K_{M,AmpG} + [\text{Disacch}_{P,peri}]} \tag{ii}$$

$$\frac{d[\text{Disacch}_{P,peri}]}{dt} = J_{Slt} - J_{AmpG} - \mu[\text{Disacch}_{P,peri}] \tag{14}$$

When the lytic enzymes (**Slt**) degrade uncrosslinked peptidoglycan (PG_{UP}), the resulting debris are pentapeptide-carrying disaccharides (referred to as muropeptides). To prevent toxic accumulation and trigger recycling, these muropeptides are actively pumped back into the cytoplasm by the inner-membrane permease AmpG. This AmpG transport step is critical, as this specific breakdown product act as the signaling molecules that trigger **AmpC beta-lactamase resistance**.

2.2.6 Periplasmic Tetrapeptide Disaccharide ($Disacch_{T,peri}$) - NAG-1,6-anh-NAM-tetrapeptide

$$J_{Slt} = \frac{V_{max,Slt}[\text{PG}_{UT}]}{K_{M,Slt} + [\text{PG}_{UT}]} \tag{i}$$

$$J_{AmpG} = \frac{V_{max,AmpG}[\text{Disacch}_{T,peri}]}{K_{M,AmpG} + [\text{Disacch}_{T,peri}]} \tag{ii}$$

$$\frac{d[\text{Disacch}_{T,peri}]}{dt} = J_{Slt} - J_{AmpG} - \mu[\text{Disacch}_{T,peri}] \tag{15}$$

This mirrors equation 14, but applies to the breakdown debris of the uncrosslinked tetrapeptide-harboring peptidoglycan (PG_{UT}). When degraded

by Slt, they yield tetrapeptide-carrying disaccharides. Like their pentapeptide counterparts, they are transported back into the cytoplasm via the AmpG permease for recycling. They do not contribute to AmpC induction.

2.3 Peptidoglycan Recycling (Cytoplasmic Reactions)

The following equations model the **peptidoglycan recycling pathway** inside the bacterial cytoplasm. Once the periplasmic breakdown products (muropeptides) are pumped into the cell by AmpG, a network of enzymes (**NagZ, AmpD, LdcA**) systematically dismantles them into reusable building blocks.

This specific pathway is highly critical because the accumulation of intermediate breakdown products (specifically the anhydro-muropeptide in its pentapeptide form) triggers the massive overproduction of the AmpC beta-lactamase.

2.3.1 Cytoplasmic Pentapeptide Disaccharide ($Disacch_{P,cyto}$) - NAG-1,6-anh-NAM-pentapeptide

$$J_{AmpG,P} = \left(\frac{V_{peri}}{V_{cyto}}\right) \cdot \left(\frac{V_{max,AmpG} \cdot [\text{Disacch}_{P,peri}]}{K_{M,AmpG} + [\text{Disacch}_{P,peri}]}\right) \quad \text{(i)}$$

$$J_{NagZ,P} = \frac{V_{max,NagZ,Disacch_P} \cdot [\text{Disacch}_{P,cyto}]}{K_{M,NagZ,Disacch_P} + [\text{Disacch}_{P,cyto}]} \quad \text{(ii)}$$

$$J_{AmpD,P} = \frac{V_{max,AmpD,Disacch_P} \cdot [\text{Disacch}_{P,cyto}]}{K_{M,AmpD,Disacch_P} + [\text{Disacch}_{P,cyto}]} \quad \text{(iii)}$$

$$\frac{d[\text{Disacch}_{P,cyto}]}{dt} = J_{AmpG,P} - J_{NagZ,P} - J_{AmpD,P} - \mu[\text{Disacch}_{P,cyto}] \quad (16)$$

This equation tracks the entry and initial processing of pentapeptide disaccharides.

- **Influx:** They enter from the periplasm via the **AmpG transporter** (scaled by the volume difference between the periplasm and cytoplasm).
- **Consumption:** Once inside, they face a fork in the road. They are degraded either by **NagZ** (which cleaves off the NAG sugar to form Anh_P, the monosaccharide 1,6-anh-NAM-pentapeptide) or by **AmpD** (an amidase that cleaves off the entire pentapeptide stem).

2.3.2 Cytoplasmic Tetrapeptide Disaccharides ($Disacch_{T,cyto}$) - NAG-1,6-anh-NAM-tetrapeptide

$$J_{AmpG,T} = \left(\frac{V_{peri}}{V_{cyto}}\right) \cdot \left(\frac{V_{max,AmpG} \cdot [\text{Disacch}_{T,peri}]}{K_{M,AmpG} + [\text{Disacch}_{T,peri}]}\right) \quad \text{(i)}$$

$$J_{NagZ,T} = \frac{V_{max,NagZ,Disacch_T} \cdot [\text{Disacch}_{T,cyto}]}{K_{M,NagZ,Disacch_T} + [\text{Disacch}_{T,cyto}]} \quad \text{(ii)}$$

$$J_{AmpD,T} = \frac{V_{max,AmpD,Disacch_T} \cdot [\text{Disacch}_{T,cyto}]}{K_{M,AmpD,Disacch_T} + [\text{Disacch}_{T,cyto}]} \quad \text{(iii)}$$

$$J_{LdcA,T} = \frac{V_{max,LdcA,Disacch_T} \cdot [\text{Disacch}_{T,cyto}]}{K_{M,LdcA,Disacch_T} + [\text{Disacch}_{T,cyto}]} \quad \text{(iv)}$$

$$\frac{d[\text{Disacch}_{T,cyto}]}{dt} = J_{AmpG,T} - J_{NagZ,T} - J_{AmpD,T} - J_{LdcA,T} - \mu[\text{Disacch}_{T,cyto}] \quad (17)$$

Similar to equation 16, this tracks the tetrapeptide variants entering via **AmpG**.

It faces a three-way fork. It can be processed by **NagZ** (removing NAG) or **AmpD** (removing the peptide). Additionally, it can be processed by **LdcA**, an enzyme that trims the terminal D-Alanine amino acid off the tetrapeptide to create a tripeptide ($Disacch_{Tri}$).

2.3.3 Cytoplasmic Tripeptide Disaccharides ($Disacch_{Tri}$) - NAG-1,6-anh-NAM-tripeptide

$$J_{LdcA,T} = \frac{V_{max,LdcA,Disacch_T} \cdot [\text{Disacch}_{T,cyto}]}{K_{M,LdcA,Disacch_T} + [\text{Disacch}_{T,cyto}]} \quad \text{(i)}$$

$$J_{NagZ,Tri} = \frac{V_{max,NagZ,Disacch_{Tri}} \cdot [\text{Disacch}_{Tri}]}{K_{M,NagZ,Disacch_{Tri}} + [\text{Disacch}_{Tri}]} \quad \text{(ii)}$$

$$J_{AmpD,Tri} = \frac{V_{max,AmpD,Disacch_{Tri}} \cdot [\text{Disacch}_{Tri}]}{K_{M,AmpD,Disacch_{Tri}} + [\text{Disacch}_{Tri}]} \quad \text{(iii)}$$

$$\frac{d[\text{Disacch}_{Tri}]}{dt} = J_{LdcA,T} - J_{NagZ,Tri} - J_{AmpD,Tri} - \mu[\text{Disacch}_{Tri}] \quad (18)$$

This intermediate is formed entirely by the trimming action of **LdcA** on the tetrapeptide (the positive term). It is then further degraded by either **NagZ** or **AmpD**.

2.3.4 Peptide-free Disaccharides ($Disacch$) - NAG-1,6-anh-NAM

$$J_{AmpD,P} = \frac{V_{max,AmpD,Disacch_P} \cdot [\text{Disacch}_{P,cyto}]}{K_{M,AmpD,Disacch_P} + [\text{Disacch}_{P,cyto}]} \quad \text{(i)}$$

$$J_{AmpD,T} = \frac{V_{max,AmpD,Disacch_T} \cdot [\text{Disacch}_{T,cyto}]}{K_{M,AmpD,Disacch_T} + [\text{Disacch}_{T,cyto}]} \quad \text{(ii)}$$

$$J_{AmpD,Tri} = \frac{V_{max,AmpD,Disacch_{Tri}} \cdot [\text{Disacch}_{Tri}]}{K_{M,AmpD,Disacch_{Tri}} + [\text{Disacch}_{Tri}]} \quad \text{(iii)}$$

$$J_{NagZ,Disacch} = \frac{V_{max,NagZ,Disacch} \cdot [\text{Disacch}]}{K_{M,NagZ,Disacch} + [\text{Disacch}]} \quad \text{(iv)}$$

$$\frac{d[\text{Disacch}]}{dt} = J_{AmpD,P} + J_{AmpD,T} + J_{AmpD,Tri} - J_{NagZ,Disacch} - \mu[\text{Disacch}] \tag{19}$$

If **AmpD** acts on the disaccharides *before* NagZ does, it strips away the peptide stem (whether penta, tetra, or tri) leaving just the naked NAG-NAM disaccharide. This equation pools those three AmpD reactions into synthesis terms, and models the subsequent consumption of this bare disaccharide by **NagZ** (which will split the two sugars).

2.3.5 1,6-anh-NAM-pentapeptide (Anh_P) - AmpR Activating Muropeptide

$$J_{NagZ,P} = \frac{V_{max,NagZ,Disacch_P} \cdot [\text{Disacch}_{P,cyto}]}{K_{M,NagZ,Disacch_P} + [\text{Disacch}_{P,cyto}]} \quad \text{(i)}$$

$$J_{AmpD,AnhP} = \frac{V_{max,AmpD,P} \cdot [\text{Anh}_P]}{K_{M,AmpD,P} + [\text{Anh}_P]} \quad \text{(ii)}$$

$$\frac{d[\text{Anh}_P]}{dt} = J_{NagZ,P} - J_{AmpD,AnhP} - \mu[\text{Anh}_P] \tag{20}$$

This is formed when **NagZ** cleaves the NAG sugar off the pentapeptide disaccharide. This leaves a 1,6-anhydro-NAM ring attached to a pentapeptide. It is subsequently degraded by **AmpD** (which will strip the peptide off). This is the **AmpR activating muropeptide**, which leads to **AmpC induction**.

2.3.6 1,6-anh-NAM-tetrapeptide (Anh_T)

$$J_{NagZ,T} = \frac{V_{max,NagZ,Disacch_T} \cdot [\text{Disacch}_{T,cyto}]}{K_{M,NagZ,Disacch_T} + [\text{Disacch}_{T,cyto}]} \quad \text{(i)}$$

$$J_{AmpD,AnhT} = \frac{V_{max,AmpD,T} \cdot [\text{Anh}_T]}{K_{M,AmpD,T} + [\text{Anh}_T]} \quad \text{(ii)}$$

$$J_{LdcA,AnhT} = \frac{V_{max,LdcA} \cdot [\text{Anh}_T]}{K_{M,AmpD,T} + [\text{Anh}_T]} \quad \text{(iii)}$$

$$\frac{d[\text{Anh}_T]}{dt} = J_{NagZ,T} - J_{AmpD,AnhT} - J_{LdcA,AnhT} - \mu[\text{Anh}_T] \quad (21)$$

Formed when **NagZ** acts on the tetrapeptide disaccharide. It is consumed by **AmpD** (stripping the peptide) or trimmed into a tripeptide by **LdcA**.

2.3.7 1,6-anh-NAM-tripeptide (Anh_{Tri})

$$J_{NagZ,Tri} = \frac{V_{max,NagZ,Disacch_{Tri}} \cdot [\text{Disacch}_{Tri}]}{K_{M,NagZ,Disacch_{Tri}} + [\text{Disacch}_{Tri}]} \quad \text{(i)}$$

$$J_{LdcA,AnhT} = \frac{V_{max,LdcA} \cdot [\text{Anh}_T]}{K_{M,LdcA} + [\text{Anh}_T]} \quad \text{(ii)}$$

$$J_{AmpD,AnhTri} = \frac{V_{max,AmpD,Tri} \cdot [\text{Anh}_{Tri}]}{K_{M,AmpD,Tri} + [\text{Anh}_{Tri}]} \quad \text{(iii)}$$

$$\frac{d[\text{Anh}_{Tri}]}{dt} = J_{NagZ,Tri} + J_{LdcA,AnhT} - J_{AmpD,AnhTri} - \mu[\text{Anh}_{Tri}] \quad (22)$$

Formed either via **NagZ** acting on the tripeptide disaccharide, or **LdcA** trimming (Anh_T). It is eventually degraded by **AmpD**.

For a long time, this muropeptide was thought to be one of the primary AmpR-activating muropeptides, but Dik et al. (2017) showed that only Anh_P acts as the activator for AmpR.

2.3.8 1,6-anh-NAM ($anhNAM$)

$$J_{AmpD,AnhP} = \frac{V_{max,AmpD,P} \cdot [\text{Anh}_P]}{K_{M,AmpD,P} + [\text{Anh}_P]} \tag{i}$$

$$J_{AmpD,AnhT} = \frac{V_{max,AmpD,T} \cdot [\text{Anh}_T]}{K_{M,AmpD,T} + [\text{Anh}_T]} \tag{ii}$$

$$J_{AmpD,AnhTri} = \frac{V_{max,AmpD,Tri} \cdot [\text{Anh}_{Tri}]}{K_{M,AmpD,Tri} + [\text{Anh}_{Tri}]} \tag{iii}$$

$$J_{NagZ,Disacch} = \frac{V_{max,NagZ,Disacch} \cdot [\text{Disacch}]}{K_{M,NagZ,Disacch} + [\text{Disacch}]} \tag{iv}$$

$$J_{AnmK} = \frac{V_{max,AnmK} \cdot [\text{anhNAM}]}{K_{M,AnmK} + [\text{anhNAM}]} \tag{v}$$

$$\frac{d[\text{anhNAM}]}{dt} = J_{AmpD,AnhP} + J_{AmpD,AnhT} + J_{AmpD,AnhTri} + J_{NagZ,Disacch} - J_{AnmK} - \mu[\text{anhNAM}] \tag{23}$$

This is the convergence point of the degradation pathways. Whether AmpD acted first or NagZ acted first, all muropeptides are eventually stripped of both their peptide stem and their NAG sugar, leaving bare **anhNAM**. This molecule is then consumed/phosphorylated by the kinase **AnmK**.

2.3.9 NAM-6-Phosphate (NAM-6-P)

$$J_{AnmK} = \frac{V_{max,AnmK} \cdot [\text{anhNAM}]}{K_{M,AnmK} + [\text{anhNAM}]} \tag{i}$$

$$J_{MurQ} = \frac{V_{max,MurQ} \cdot [\text{NAM-6-P}]}{K_{M,MurQ} + [\text{NAM-6-P}]} \tag{ii}$$

$$\frac{d[\text{NAM-6-P}]}{dt} = J_{AnmK} - J_{MurQ} - \mu[\text{NAM-6-P}] \tag{24}$$

This equation tracks cytosolic NAM-6-P, produced via **AnmK** phosphorylation of anhNAM (J_{AnmK}) and consumed by **MurQ** (J_{MurQ}) into NAG-6-P and D-lactate to feed central carbohydrate metabolism.

2.3.10 NAG-6-Phosphate ($NAG-6-P$)

$$J_{MurQ} = \frac{V_{max,MurQ} \cdot [\text{NAM-6-P}]}{K_{M,MurQ} + [\text{NAM-6-P}]} \tag{i}$$

$$J_{NagK} = \frac{V_{max,NagK} \cdot [\text{NAG}]}{K_{M,NagK} + [\text{NAG}]} \tag{ii}$$

$$J_{NagA} = \frac{V_{max,NagA} \cdot [\text{NAG-6-P}]}{K_{M,NagA} + [\text{NAG-6-P}]} \tag{iii}$$

$$\frac{d[\text{NAG-6-P}]}{dt} = J_{MurQ} + J_{NagK} - J_{NagA} - \mu[\text{NAG-6-P}] \tag{25}$$

This equation tracks cytosolic NAG-6-P, synthesized via **MurQ** conversion of NAM-6-P (J_{MurQ}) and **NagK** phosphorylation of free NAG (J_{NagK}), and consumed by **NagA** deacetylase (J_{NagA}).

2.3.11 Glucosamine 6-phosphate ($GlcN-6-P$)

$$J_{NagA} = \frac{V_{max,NagA} \cdot [\text{NAG-6-P}]}{K_{M,NagA} + [\text{NAG-6-P}]} \tag{i}$$

$$J_{GlmS} = V_{max,GlmS} \cdot \left(\frac{[\text{Fru-6-P}]_{basal}}{K_{M,GlmS,F6P} + [\text{Fru-6-P}]_{basal}} \right) \cdot \left(\frac{[\text{Gln}]_{basal}}{K_{M,GlmS,Gln} + [\text{Gln}]_{basal}} \right) \tag{ii}$$

$$J_{NagB} = \frac{V_{max,NagB} \cdot [\text{GlcN-6-P}]}{K_{M,NagB} + [\text{GlcN-6-P}]} \tag{iii}$$

$$J_{GlmM} = \frac{V^{fwd}_{max,GlmM} \cdot \frac{[\text{GlcN-6-P}]}{K_{M,GlmM,6P}} - V^{rev}_{max,GlmM} \cdot \frac{[\text{GlcN-1-P}]}{K_{M,GlmM,1P}}}{1 + \frac{[\text{GlcN-6-P}]}{K_{M,GlmM,6P}} + \frac{[\text{GlcN-1-P}]}{K_{M,GlmM,1P}}} \tag{iv}$$

$$\frac{d[\text{GlcN-6-P}]}{dt} = J_{NagA} + J_{GlmS} - J_{NagB} - J_{GlmM} - \mu[\text{GlcN-6-P}] \tag{26}$$

This equation tracks cytosolic GlcN-6-P, synthesized via **NagA** deacetylation (J_{NagA}) and *de novo* **GlmS** synthase (J_{GlmS}), and consumed by **NagB** deaminase (J_{NagB}) and reversible **GlmM** mutase isomerization (J_{GlmM}).

2.3.12 Glucosamine 1-phosphate ($GlcN-1-P$)

$$J_{GlmM} = \frac{V^{fwd}_{max,GlmM} \cdot \frac{[\text{GlcN-6-P}]}{K_{M,GlmM,6P}} - V^{rev}_{max,GlmM} \cdot \frac{[\text{GlcN-1-P}]}{K_{M,GlmM,1P}}}{1 + \frac{[\text{GlcN-6-P}]}{K_{M,GlmM,6P}} + \frac{[\text{GlcN-1-P}]}{K_{M,GlmM,1P}}} \quad \text{(i)}$$

$$J_{GlmU} = \frac{V_{max,GlmU} \cdot [\text{GlcN-1-P}]}{K_{M,GlmU} + [\text{GlcN-1-P}]} \quad \text{(ii)}$$

$$\frac{d[\text{GlcN-1-P}]}{dt} = J_{GlmM} - J_{GlmU} - \mu[\text{GlcN-1-P}] \quad (27)$$

This equation tracks cytosolic glucosamine 1-phosphate (GlcN-1-P). Its synthesis is primarily driven by **GlmM** (J_{GlmM}), a phosphoglucosamine mutase operating via reversible, two-substrate Michaelis-Menten kinetics. The net flux direction and rate depend directly on the competitive saturation between GlcN-6-P and GlcN-1-P, dynamically shifting the phosphate group between the C6 and C1 positions to maintain thermodynamic equilibrium. GlcN-1-P is subsequently consumed by **GlmU** (J_{GlmU}) to initiate nucleotide-sugar synthesis.

2.3.13 N-Acetyl Glucosamine (NAG)

$$J_{NagZ,Disacch} = \frac{V_{max,NagZ,Disacch} \cdot [\text{Disacch}]}{K_{M,NagZ,Disacch} + [\text{Disacch}]} \quad \text{(i)}$$

$$J_{NagZ,P} = \frac{V_{max,NagZ,Disacch_P} \cdot [\text{Disacch}_{P,cyto}]}{K_{M,NagZ,Disacch_P} + [\text{Disacch}_{P,cyto}]} \quad \text{(ii)}$$

$$J_{NagZ,T} = \frac{V_{max,NagZ,Disacch_T} \cdot [\text{Disacch}_{T,cyto}]}{K_{M,NagZ,Disacch_T} + [\text{Disacch}_{T,cyto}]} \quad \text{(iii)}$$

$$J_{NagZ,Tri} = \frac{V_{max,NagZ,Disacch_{Tri}} \cdot [\text{Disacch}_{Tri}]}{K_{M,NagZ,Disacch_{Tri}} + [\text{Disacch}_{Tri}]} \quad \text{(iv)}$$

$$J_{NagK} = \frac{V_{max,NagK} \cdot [\text{NAG}]}{K_{M,NagK} + [\text{NAG}]} \quad \text{(v)}$$

$$\frac{d[\text{NAG}]}{dt} = J_{NagZ,Disacch} + J_{NagZ,P} + J_{NagZ,T} + J_{NagZ,Tri} - J_{NagK} - \mu[\text{NAG}] \quad (28)$$

This equation tracks the cytosolic pool of free N-acetylglucosamine (NAG). It accumulates as a universal byproduct of **NagZ** β-N-acetylglucosaminidase

activity, which hydrolyzes the glycosidic bond of incoming muropeptides to yield NAG and the corresponding anhydro-NAM derivatives. The synthesis fluxes account for all NagZ substrates: peptide-free disaccharides ($J_{NagZ,Disacch}$), pentapeptides ($J_{NagZ,P}$), tetrapeptides ($J_{NagZ,T}$), and tripeptides ($J_{NagZ,Tri}$). This free NAG pool is then salvaged through phosphorylation by the kinase **NagK** (J_{NagK}) to form NAG-6-P.

2.3.14 Free Pentapeptide Stem ($Stem_{Penta}$)

$$J_{AmpD,P} = \frac{V_{max,AmpD,Disacch_P} \cdot [\text{Disacch}_{P,cyto}]}{K_{M,AmpD,Disacch_P} + [\text{Disacch}_{P,cyto}]} \quad \text{(i)}$$

$$J_{AmpD,AnhP} = \frac{V_{max,AmpD,P} \cdot [\text{Anh}_P]}{K_{M,AmpD,P} + [\text{Anh}_P]} \quad \text{(ii)}$$

$$J_{Mpl,Penta} = V_{max,Mpl,Penta} \cdot \left(\frac{[\text{UDP-NAM}]}{K_{M,Mpl,NAM} + [\text{UDP-NAM}]} \right) \cdot \left(\frac{[\text{Stem}_{Penta}]}{K_{M,Mpl,Penta} + [\text{Stem}_{Penta}]} \right) \quad \text{(iii)}$$

$$\frac{d[\text{Stem}_{Penta}]}{dt} = J_{AmpD,P} + J_{AmpD,AnhP} - J_{Mpl,Penta} - \mu[\text{Stem}_{Penta}] \quad (29)$$

This mass-balance equation tracks the cytosolic pool of free pentapeptide stems (Stem_{Penta}). These stems are generated via **AmpD** amidase cleavage, which strips them from incoming disaccharide pentapeptides ($J_{AmpD,P}$) and anhydro-NAM pentapeptides ($J_{AmpD,AnhP}$). The free stems are then directly salvaged by the ligase **Mpl** ($J_{Mpl,Penta}$). Operating via bi-substrate Michaelis-Menten kinetics, Mpl attaches the intact pentapeptide onto UDP-NAM, efficiently bypassing the ATP-intensive *de novo* sequential peptide addition.

2.3.15 Free Tetrapeptide Stem ($Stem_{Tetra}$)

$$J_{AmpD,T} = \frac{V_{max,AmpD,Disacch_T} \cdot [\text{Disacch}_{T,cyto}]}{K_{M,AmpD,Disacch_T} + [\text{Disacch}_{T,cyto}]} \quad \text{(i)}$$

$$J_{AmpD,AnhT} = \frac{V_{max,AmpD,T} \cdot [\text{Anh}_T]}{K_{M,AmpD,T} + [\text{Anh}_T]} \quad \text{(ii)}$$

$$J_{LdcA,Stem} = \frac{V_{max,LdcA,stem} \cdot [\text{Stem}_{Tetra}]}{K_{M,LdcA,stem} + [\text{Stem}_{Tetra}]} \quad \text{(iii)}$$

$$\frac{d[\text{Stem}_{Tetra}]}{dt} = J_{AmpD,T} + J_{AmpD,AnhT} - J_{LdcA,Stem} - \mu[\text{Stem}_{Tetra}] \quad (30)$$

This equation tracks the cytosolic pool of free tetrapeptide stems (Stem_{Tetra}). These peptide chains accumulate via the amidase activity of **AmpD**, which cleaves them from incoming disaccharide tetrapeptides ($J_{AmpD,T}$) and anhydro-NAM tetrapeptides ($J_{AmpD,AnhT}$). Unlike pentapeptides, free tetrapeptides are not directly recycled by Mpl; instead, they are consumed by the LD-carboxypeptidase **LdcA** ($J_{LdcA,Stem}$). LdcA trims the terminal D-alanine to yield free tripeptide stems, directing them back into the salvage pathway.

2.3.16 Free Tripeptide Stem ($Stem_{Tri}$)

$$J_{AmpD,Tri} = \frac{V_{max,AmpD,Disacch_{Tri}} \cdot [\text{Disacch}_{Tri}]}{K_{M,AmpD,Disacch_{Tri}} + [\text{Disacch}_{Tri}]} \quad \text{(i)}$$

$$J_{AmpD,AnhTri} = \frac{V_{max,AmpD,Tri} \cdot [\text{Anh}_{Tri}]}{K_{M,AmpD,Tri} + [\text{Anh}_{Tri}]} \quad \text{(ii)}$$

$$J_{LdcA,Stem} = \frac{V_{max,LdcA,stem} \cdot [\text{Stem}_{Tetra}]}{K_{M,LdcA,stem} + [\text{Stem}_{Tetra}]} \quad \text{(iii)}$$

$$J_{Mpl,Tri} = V_{max,Mpl,Tri} \cdot \left(\frac{[\text{UDP-NAM}]}{K_{M,Mpl,NAM} + [\text{UDP-NAM}]}\right) \cdot \left(\frac{[\text{Stem}_{Tri}]}{K_{M,Mpl,Tri} + [\text{Stem}_{Tri}]}\right) \quad \text{(iv)}$$

$$\frac{d[\text{Stem}_{Tri}]}{dt} = J_{AmpD,Tri} + J_{AmpD,AnhTri} + J_{LdcA,Stem} - J_{Mpl,Tri} - \mu[\text{Stem}_{Tri}] \quad (31)$$

This equation tracks the cytosolic pool of free tripeptide stems (Stem_{Tri}). These stems accumulate via three distinct synthesis fluxes: **AmpD** amidase cleavage of incoming disaccharide tripeptides ($J_{AmpD,Tri}$) and anhydro-NAM tripeptides ($J_{AmpD,AnhTri}$), as well as **LdcA** carboxypeptidase trimming of the terminal D-alanine from free tetrapeptide stems ($J_{LdcA,Stem}$). The tripeptides are then salvaged by the ligase **Mpl** ($J_{Mpl,Tri}$). Operating via bi-substrate Michaelis-Menten kinetics (split across two lines for readability), Mpl attaches the tripeptide directly onto UDP-NAM, bypassing the ATP-intensive MurC, MurD, and MurE *de novo* synthesis steps

2.3.17 Und-P Recycling Reactions

The following three equations model the recycling of the **lipid carrier** (Undecaprenyl phosphate, or Und-P), which acts like a biological conveyor belt.

Because bacteria only have a very limited pool of these lipid molecules, they must be strictly recycled. After a lipid carrier transports a peptidoglycan building block across the inner membrane, it must be dephosphorylated and flipped back into the cytoplasm to pick up the next block.

2.3.18 Periplasmic Undecaprenyl Pyrophosphate ($Und-PP_{peri}$)

$$J_{RodA} = \frac{V_{max,RodA} \cdot [\text{Lipid II}_{peri}]}{K_{M,RodA} + [\text{Lipid II}_{peri}]} \tag{i}$$

$$J_{FtsW} = Z \cdot \frac{V_{max,FtsW} \cdot [\text{Lipid II}_{peri}]}{K_{M,FtsW} + [\text{Lipid II}_{peri}]} \tag{ii}$$

$$J_{PBP1a,GT} = \frac{k_{cat,PBP1a,GT}[PBP1a][\text{Lipid II}_{peri}]}{K_{M,PBP1a,GT} + [\text{Lipid II}_{peri}]} \tag{iii}$$

$$J_{PBP1b,GT} = Z \cdot \frac{k_{cat,PBP1b,GT}[PBP1b][\text{Lipid II}_{peri}]}{K_{M,PBP1b,GT} + [\text{Lipid II}_{peri}]} \tag{iv}$$

$$J_{BacA} = \frac{V_{max,BacA} \cdot [\text{Und-PP}_{peri}]}{K_{M,BacA} + [\text{Und-PP}_{peri}]} \tag{v}$$

$$\frac{d[\text{Und-PP}_{peri}]}{dt} = v_{de_novo} + J_{RodA} + J_{FtsW} + J_{PBP1a,GT} + J_{PBP1b,GT} - J_{BacA} - \mu[\text{Und-PP}_{peri}] \tag{32}$$

This equation tracks the lipid carrier the exact moment it drops off its cargo on the periplasmic side of the membrane.

- **Influx/Release:** When the glycosyltransferases (**RodA** for elongation, **FtsW** for division) polymerize Lipid II into the cell wall, they sever the bond between the sugar unit and the lipid carrier. Because Lipid II has a double-phosphate linkage, the released carrier is in the inactive pyrophosphate form ($Und-PP$).
- v_{de_novo}: There is also a small basal rate of fresh, newly synthesized carrier molecules entering the system here from the *de novo* lipid carrier synthesis pathway.

- **Consumption:** To be reused, **Und-PP** must lose one phosphate group. It is consumed by the integral membrane phosphatase **BacA**, which cleaves off the extra phosphate.

2.3.19 Periplasmic Undecaprenyl Phosphate ($Und - P_{peri}$)

$$J_{BacA} = \frac{V_{max,BacA} \cdot [\text{Und-PP}_{peri}]}{K_{M,BacA} + [\text{Und-PP}_{peri}]} \tag{i}$$

$$J_{DedA} = \frac{V_{max,DedA} \cdot [\text{Und-P}_{peri}]}{K_{M,DedA} + [\text{Und-P}_{peri}]} \tag{ii}$$

$$\frac{d[\text{Und-P}_{peri}]}{dt} = J_{BacA} - J_{DedA} - \mu[\text{Und-P}_{peri}] \tag{33}$$

This equation tracks the newly activated carrier, which is now in its active monophosphate form ($Und-P$) but is stuck on the wrong side of the inner membrane (the periplasmic face).

- **Synthesis:** It is produced entirely by the dephosphorylation activity of **BacA** (matching the consumption term from the previous equation).
- **Consumption:** To be reused, it must be physically flipped back across the inner membrane to face the inside of the cell. This transport is catalyzed by a flippase protein, modeled here as a member of the DedA family (or similar Und-P transporters like UptA), following Michaelis-Menten kinetics.

2.3.20 Cytoplasmic Undecaprenyl Phosphate ($Und - P_{cyto}$)

$$J_{DedA} = \left(\frac{V_{peri}}{V_{cyto}}\right) \cdot \left(\frac{V_{max,DedA} \cdot [\text{Und-P}_{peri}]}{K_{M,DedA} + [\text{Und-P}_{peri}]}\right) \tag{i}$$

$$J_{MraY} = V_{max,MraY} \cdot \left(\frac{[\text{UDP-NAM-pentapeptide}]}{K_{M,UDP-NAM-pentapeptide} + [\text{UDP-NAM-pentapeptide}]}\right) \cdot \left(\frac{[\text{Und-P}_{cyto}]}{K_{M,UndP} + [\text{Und-P}_{cyto}]}\right) \tag{ii}$$

$$\frac{d[\text{Und-P}_{cyto}]}{dt} = J_{DedA} - J_{MraY} - \mu[\text{Und-P}_{cyto}] \tag{34}$$

This final equation closes the entire peptidoglycan biosynthesis loop. The empty, active lipid carrier is now back in the cytoplasm, ready to pick up a new building block.

- **Influx:** It enters the cytoplasm via the **DedA** flippase. Just like the AmpG transporter equation earlier, this term is multiplied by $(\frac{V_{peri}}{V_{cyto}})$ to account for the change in concentration when moving from the small volume of the periplasm to the much larger volume of the cytoplasm.
- **Consumption:** The carrier is consumed by the membrane enzyme **MraY**, which attaches a fresh UDP-NAM-pentapeptide to it to form Lipid I. (In equation 8, this exact bi-substrate MraY term acting as the synthesis rate for Lipid I can be seen, thereby perfectly completing the cycle).

2.4 AmpR Regulation and AmpC Production

The following equations represent the genetic regulation and production of AmpC, the primary beta-lactamase enzyme. The model captures the molecular tug-of-war for the transcription factor AmpR. AmpR controls the expression of the *ampC* gene and can bind to three competing molecules: two **activators** (the cell wall breakdown products, Anh_P and $Disacch_{P,cyto}$) and a **repressor** (the cell wall building block, UDP-NAM-pentapeptide).

2.4.1 AmpR transcription and translation

$$[\text{AmpR}_{total}] = [\text{AmpR}] \cdot \left[1 + \left(\frac{[\text{UDP-NAM-pentapeptide}]}{K_R}\right)^n + \left(\frac{[\text{Anh}_P]}{K_{A1}}\right)^n + \left(\frac{[\text{Disacch}_{P,cyto}]}{K_{A2}}\right)^n\right] \quad \text{(i)}$$

This algebraic equation partitions the total regulator pool across four conformational states: the free (apo) form $[AmpR]$, and three ligand-bound forms generated by competitive equilibrium binding to the repressor (UDP-MurNAc-pentapeptide, dissociation constant K_R) and two activating muropeptides (Anh_P and $Disacch_{P,cyto}$, constants K_{A1} and K_{A2}). The shared Hill exponent n reflects AmpR's tetrameric cooperativity and is treated as a single protein-level property rather than a ligand-specific parameter, since it

governs the stoichiometry of the oligomer rather than the chemistry of any one ligand.

$$\frac{d[\mathrm{mRNA}_{AmpR}]}{dt} = N_{copy} \cdot v_{tx,AmpR} \cdot \left\{ \frac{1}{1 + \left(\frac{[\mathrm{AmpR}_{total}]}{K_{autoR}} \right)^{n_{autoR}}} \right\} - (k_{deg,mRNA,AmpR} + \mu) \cdot [\mathrm{mRNA}_{AmpR}] \tag{35}$$

N_{copy} or the copy number of the plasmid accounts for the plasmid-borne (multi-copy) origin of the *Citrobacter freundii* AmpR gene. The autorepression term is keyed to $[AmpR_{total}]$ rather than to any individual bound state, consistent with biochemical evidence that AmpR occupies its own promoter irrespective of which ligand, if any, it is currently bound to, i.e., the negative feedback is constitutive and ligand-state-independent. Loss is first-order, combining transcript degradation and growth dilution.

$$\frac{d[AmpR_{total}]}{dt} = k_{tl,AmpR} \cdot [mRNA_{AmpR}] - (k_{deg,p,AmpR} + \mu) \cdot [AmpR_{total}] \tag{36}$$

Newly translated protein enters the free pool directly; degradation and dilution act on the free form only, with the distribution across bound states recovered at each instant from the algebraic partitioning relation above rather than tracked through separate kinetic equations, i.e., ligand binding is assumed to equilibrate rapidly relative to the timescales of transcription, translation, and dilution.

2.4.2 AmpC transcription and translation

$$\frac{d[\mathrm{mRNA}_{AmpC}]}{dt} = N_{copy} \cdot \left[k_{tx,rep} \left(\frac{[\text{UDP-NAM-pentapeptide}]}{K_R} \right)^n + k_{tx,basal} + k_{tx,act} \left\{ \left(\frac{[\mathrm{Anh}_P]}{K_{A1}} \right)^n + \left(\frac{[\mathrm{Disacch}_{P,cyto}]}{K_{A2}} \right)^n \right\} \right] \cdot [\mathrm{AmpR}] - (k_{deg,mRNA,AmpC} + \mu) \cdot [\mathrm{mRNA}_{AmpC}] \tag{37}$$

Each bracketed term corresponds to one of the three ligand-defined AmpR states and carries its own per-molecule transcription rate: $k_{tx,rep}$ for the repressor-bound state, $k_{tx,basal}$ for the unliganded (apo) state, and a shared $k_{tx,act}$ for both activator-bound states, reflecting the assumption that Anh_P and $\mathrm{Disacch}_{P,cyto}$ differ in binding affinity (K_{A1} vs. K_{A2}) but not in the

magnitude of the transcriptional output once bound. Multiplying the whole bracket by [AmpR] (rather than by [AmpR_{total}]) ties transcriptional output specifically to the free/apo pool, consistent with the equilibrium-partitioning relation defined in the AmpR module: each state's absolute concentration is recovered as [AmpR] times its own ligand-ratio term, so the sum correctly reduces to per-state $k_{tx}\times$ [state concentration] without requiring a separate normalization. N_{copy} again reflects the plasmid-borne, multi-copy origin of the gene. Loss is first-order (transcript degradation plus growth dilution).

$$\frac{d[\text{AmpC}]}{dt} = \left(\frac{V_{cyto}}{V_{peri}}\right) \cdot k_{tl} \cdot [\text{mRNA}_{AmpC}] - (k_{deg,p,AmpC} + \mu) \cdot [\text{AmpC}] \quad (38)$$

The volume-ratio factor V_{cyto}/V_{peri} converts the translation flux, which occurs in the cytoplasm, into the concentration units of the periplasmic compartment in which mature AmpC resides following signal-peptide-directed export: a correction not applied to AmpR, which remains cytoplasmic to perform its DNA-binding function. Degradation and dilution act on the periplasmic (exported) pool.

2.5 β-Lactam and PBP Kinetics

The following equations describe antibiotic dynamics and kinetics: how the drug accumulates inside the bacterial periplasm, and how it neutralizes its cellular targets (the PBPs).

2.5.1 β-Lactam

$$\begin{aligned}\frac{d[\text{BL}]}{dt} = k_{diff,BL} \cdot ([\text{BL}]_{ext} - [\text{BL}]) - \frac{k_{cat,AmpC} \cdot [\text{AmpC}] \cdot [\text{BL}]}{K_{M,AmpC} + [\text{BL}]} - \frac{V_{max,eff} \cdot [\text{BL}]}{K_{M,eff} + [\text{BL}]} \\ - \Big(k_{acyl,PBP1a} \cdot [\text{PBP1a}] + k_{acyl,PBP1b} \cdot [\text{PBP1b}] + k_{acyl,PBP2} \cdot [\text{PBP2}] \\ + k_{acyl,PBP3} \cdot [\text{PBP3}] + k_{acyl,PBP4} \cdot [\text{PBP4}] + k_{acyl,PBP5} \cdot [\text{PBP5}] \\ + k_{acyl,PBP6} \cdot [\text{PBP6}] + k_{acyl,PBP7} \cdot [\text{PBP7}]\Big) \cdot [\text{BL}] - \mu[\text{BL}]\end{aligned} \quad (39)$$

Intracellular cefoxitin concentration balances passive porin-mediated influx against three loss processes: enzymatic hydrolysis by AmpC (Michaelis-Menten), efflux (structurally retained but set to zero, reflecting cefoxitin's poor AcrAB-TolC affinity), and irreversible acylation of all eight tracked PBPs, each with its own literature-derived affinity constant. The acylation term simultaneously depletes free drug and inactivates cell-wall synthesis

machinery via the corresponding PBP equations, coupling drug kinetics directly to peptidoglycan metabolism. Growth dilution acts on the free pool throughout.

2.5.2 PBPs

$$\frac{d[\text{PBP1a}]}{dt} = v_{synth,PBP1a} - k_{acyl,PBP1a} \cdot [\text{PBP1a}] \cdot [\text{BL}] - \mu[\text{PBP1a}] \quad (40)$$

$$\frac{d[\text{PBP1b}]}{dt} = v_{synth,PBP1b} - k_{acyl,PBP1b} \cdot [\text{PBP1b}] \cdot [\text{BL}] - \mu[\text{PBP1b}] \quad (41)$$

$$\frac{d[\text{PBP2}]}{dt} = v_{synth,PBP2} - k_{acyl,PBP2} \cdot [\text{PBP2}] \cdot [\text{BL}] - \mu[\text{PBP2}] \quad (42)$$

$$\frac{d[\text{PBP3}]}{dt} = v_{synth,PBP3} - k_{acyl,PBP3} \cdot [\text{PBP3}] \cdot [\text{BL}] - \mu[\text{PBP3}] \quad (43)$$

$$\frac{d[\text{PBP4}]}{dt} = v_{synth,PBP4} - k_{acyl,PBP4} \cdot [\text{PBP4}] \cdot [\text{BL}] - \mu[\text{PBP4}] \quad (44)$$

$$\frac{d[\text{PBP5}]}{dt} = v_{synth,PBP5} - k_{acyl,PBP5} \cdot [\text{PBP5}] \cdot [\text{BL}] - \mu[\text{PBP5}] \quad (45)$$

$$\frac{d[\text{PBP6}]}{dt} = v_{synth,PBP6} - k_{acyl,PBP6} \cdot [\text{PBP6}] \cdot [\text{BL}] - \mu[\text{PBP6}] \quad (46)$$

$$\frac{d[\text{PBP7}]}{dt} = v_{synth,PBP7} - k_{acyl,PBP7} \cdot [\text{PBP7}] \cdot [\text{BL}] - \mu[\text{PBP7}] \quad (47)$$

All eight PBPs (elongasome-associated PBP1a/PBP2, divisome-associated PBP1b/PBP3, and low-molecular-weight PBP4–7) follow an identical structural form: constant zeroth-order synthesis, irreversible loss to cefoxitin acylation at protein-specific rates $k_{acyl,i}$, and growth dilution. Synthesis rates are derived from proteomic abundance data; acylation constants are sourced from whole-cell competition-binding assays and span several orders of magnitude, reflecting cefoxitin's differential affinity across the PBP set (strongest for PBP4/PBP7, negligible for PBP2). This shared structure directly links drug exposure to the depletion of each synthesis enzyme, propagating into the peptidoglycan-synthesis equations in which each [PBP$_i$] appears.

2.6 FtsZ Kinetics

The following equations model the experimental manipulation of the cell using a temperature-sensitive (ts) **ftsZ84** mutant, as employed by Uehara and Park (2002), to deliberately halt cell division upon a shift to the restrictive temperature.

By inactivating the FtsZ protein, the master regulator that forms the "Z-ring" to pinch the dividing cell in half, this shift selectively shuts down the division-specific cell wall enzymes (FtsW and PBP3) that were modeled earlier.

$$Z = \Theta \cdot \frac{[\text{FtsZ}]^{n_Z}}{K^{n_Z} + [\text{FtsZ}]^{n_Z}} \tag{i}$$

Z is a Hill function of FtsZ abundance, representing the fraction of division-specific machinery (FtsW, PBP3) rendered functional at a given FtsZ level. The prefactor Θ is a time-dependent step function set to 1 under permissive conditions and 0 following the shift to the restrictive temperature, imposing the ts-*ftsZ84* mutant's abrupt, temperature-triggered loss of division competence independently of FtsZ's own protein-level dynamics.

$$\frac{d[\text{FtsZ}]}{dt} = v_{synth,FtsZ} - (k_{deg,p,FtsZ} + \mu) \cdot [\text{FtsZ}] \tag{48}$$

This equation follows standard first-order synthesis-degradation-dilution kinetics, calibrated to a literature-reported steady-state FtsZ abundance. Because the ts mutation abolishes FtsZ function rather than its expression, this equation is left unaffected by the temperature shift; the resulting phenotype is instead imposed entirely through Θ in the Z switch above.

3 Parameters

The following tables contain all the parameter values. Some were taken directly from the previous report's parameter tables:

- kinetic_parameters: This table contains literature-sourced and derived values of kinetic constants (K_M and k_cat) for all the enzymes included in the model. All enzyme abundance values are obtained from PaxDb.
- other_constants: This table contains the remaining parameter values, both literature-sourced and derived.

4 Test Conditions

We simulate four experimental conditions by toggling two control parameters in the model, matching the two-stage shift-then-drug protocol of Uehara and Park (2002):

- **ts_allele (The Division Block):** Selects the genetic background.
  - Set to 0 to simulate Wild-Type (WT), in which FtsZ remains functional throughout.
  - Set to 1 to simulate the temperature-sensitive *ftsZ84* mutant, in which the divisome switch Θ transitions from 1 to 0 upon the shift to the restrictive temperature ($t = t_{shift}$), abolishing FtsZ function without altering its synthesis.
- **BL$_{ext}$ (The Cefoxitin Dose):** Controls extracellular drug concentration, introduced only after the temperature shift ($t = t_{drug}$), reproducing the protocol's own sequence of a 30-minute shift-only period followed by drug addition.
  - Set to 0 for untreated cells.
  - Set to the strain's reported MIC (8 μg/mL) for treated cells.

4.1 The Four Experimental Conditions

WT, no BL (Baseline Control)

- **Parameters:** ts_allele = 0, BL$_{ext}$ = 0
- **Purpose:** Acts as the healthy, unstressed baseline. The model is allowed to reach steady-state homeostasis, establishing baseline levels of AmpC expression, peptidoglycan precursors, and normal PBP activity.

WT, +BL (Antibiotic Only)

- **Parameters:** ts_allele = 0, BL$_{ext}$ > 0 (set to MIC)
- **Purpose:** Simulates standard treatment of a dividing cell. Cefoxitin enters the periplasm and acylates the tracked PBP set, most strongly PBP4 and PBP7, triggering the cell-wall recycling cascade that normally results in strong induction of the AmpC resistance enzyme.

***ftsZ84*, no BL (Depletion Only)**

- **Parameters:** ts_allele = 1, BL$_{ext}$ = 0
- **Purpose:** Isolates the metabolic effect of halting cell division via the restrictive-temperature shift alone. By disabling FtsW/PBP3 activity

without antibiotic exposure, this checks whether loss of the Z-ring alone is sufficient to alter peptidoglycan precursor fluxes, independent of any drug effect.

***ftsZ84*, +BL (The Synergy Test)**

- **Parameters:** ts_allele = 1, $\mathrm{BL}_{ext} > 0$ (set to MIC)
- **Purpose:** This is the core hypothesis test, reproducing the paper's own experimental comparison. It combines FtsZ inactivation with cefoxitin treatment to determine whether the resulting bottleneck across division- and drug-affected pathways blunts AmpC induction relative to the WT response — the specific fold-change deficit ($\rho \approx 0.23$) reported by Uehara and Park.

5 Simulation Methodology

The model is implemented as a system of ordinary differential equations in Antimony/SBML, executed via the tellurium/libRoadRunner framework in Python. Each simulated condition follows a two-stage protocol matching Uehara and Park (2002): the model is first integrated to a stable steady state under permissive, drug-free conditions (30°C-equivalent baseline), representing exponentially growing cells prior to any experimental perturbation. From this shared steady state, the system branches into four parallel runs corresponding to the genotype (wild-type vs. temperature-sensitive ftsZ84) and drug-exposure (untreated vs. cefoxitin-treated) axes of the experimental design, with the restrictive-temperature shift and drug addition introduced as time-dependent step functions at their respective protocol timepoints. Steady-state convergence at each stage is verified numerically via the residual of the system's rate-of-change vector, rather than assumed from integration time alone. AmpC induction is quantified as the fold-change in AmpC concentration between paired drug-treated and untreated conditions within each genotype, directly mirroring the specific-activity ratios reported in the source literature.

6 Parameter Estimation

Since most parameters are unknown, it is more likely than not that the model will yield outputs inconsistent with literature-based expectations. The inverse problem of solving for these unknown parameter values from

the available experimental data therefore becomes not only inevitable but valuable, as it reveals the kinetic characteristics of enzymes whose catalytic profiles remain otherwise uncharacterised. We perform both manual and automated calibration to fit the parameter values against the Uehara and Park (2002) AmpC induction and repressor accumulation data.

6.1 Manual Calibration

Manual calibration proceeded by tracing flux imbalances sequentially through the model, resolving bottlenecks in Und-P recycling, peptidoglycan synthesis, and crosslinking capacity as each was uncovered, with corrections anchored where possible to independent literature values. This approach improved individual targets substantially but proved unable to satisfy the full set of experimental constraints simultaneously, as fixes to one bottleneck repeatedly exposed the next. This motivated the shift to automated, global parameter optimisation.

6.2 Automated Calibration

Manual calibration required repeatedly guessing individual parameter values and inspecting the result by eye, which does not scale once several parameters interact and several experimental targets must be matched at once. This was reformulated as a formal **optimisation problem**. Six quantities the model needed to reproduce were identified from the experimental data: two AmpC induction fold-changes (wild-type and the division-defective mutant), a genotype-parity check of AmpC induction, and three ratios describing how the repressor pool should shift across conditions, and for each, the relative error between simulated and target value was squared and combined into a single weighted score, with one additional lower-priority target (the repressor's absolute concentration, 0.2 mM (Mengin-Lecreulx et al., 1982)) included at reduced weight as a secondary check. The relative-error formulation was chosen because the targets differ enormously in scale.

$$L(\theta) = \sum_{i=1}^{6} w_i \left(\frac{y_i(\theta)}{\hat{y}_i} - 1 \right)^2 + w_{UDP} \left(\frac{[UDP\text{-}NAM\text{-}penta]_1(\theta)}{0.2} - 1 \right)^2$$

The search was carried out using **differential evolution**, a global, population-based algorithm that requires no gradient information. It was a deliberate choice, since earlier exploratory work had revealed a landscape

with sharp, cliff-like transitions and non-monotonic responses where conventional gradient-based optimisers are prone to becoming trapped. Differential evolution instead maintains a population of candidate parameter sets and improves them generation by generation purely by comparing candidates against one another, making it substantially more robust to this kind of landscape at the cost of requiring many more simulation runs to converge. A short local refinement (using **BFGS**, a gradient-based optimisation technique) step was then applied to the single best candidate the global search found, to fine-tune the result once a promising region of parameter space had already been located.

7 Testing the Calibrated Model on Different Drugs

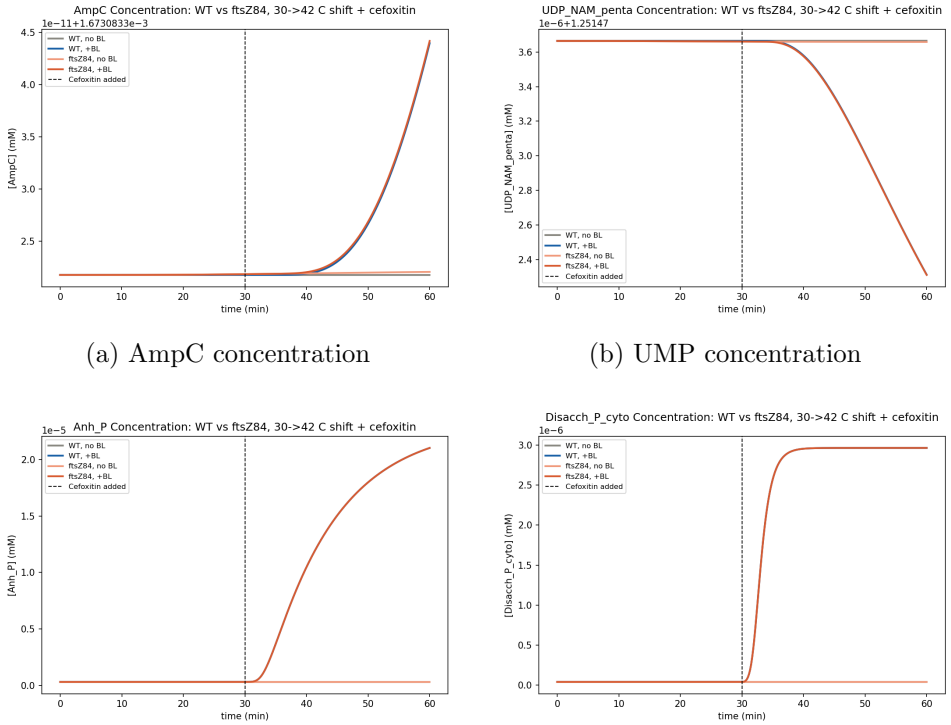
Having calibrated the model against the cefoxitin induction data reported by Uehara and Park (2002), we extend the analysis to test the model's predictions for a panel of additional β-lactams. The purpose of this exercise is to assess whether the calibrated parameter set reflects a genuine, generalisable representation of the FtsZ–AmpR–AmpC induction mechanism, or whether it instead constitutes a fit specific to cefoxitin's own pharmacological profile.

To test this, only drug-specific parameters are substituted into the calibrated model: **outer membrane permeability**, **efflux kinetics**, **PBP acylation profiles**, and **AmpC hydrolysis kinetics**. Every parameter describing the cell's own peptidoglycan biosynthesis, recycling, and AmpR–AmpC machinery is held fixed at its calibrated value. The choice of drugs tested was constrained by data availability rather than selected freely: since the validation strain expresses the *Citrobacter freundii* AmpR–AmpC construct rather than a native *E. coli* AmpC. Bauvois et al. (2005) report such kinetics for a limited set of β-lactams, and the drug panel tested here: **ampicillin**, **cephalexin**, **cefotaxime**, **cefuroxime**, and **oxacillin**, was accordingly restricted to this set.

8 Results

8.1 Simulation Results of Original, Uncalibrated Model

Although looking from Figure 1, we see visible movements in the concentrations of AmpC and UMP in the simulation timeframe, they are actually not moving at all. **AmpC and UDP-N-acetylmuramoyl-pentapeptide (UDP-NAM-penta) genuinely remain static throughout, with rel-**



(a) AmpC concentration

(b) UMP concentration

(c) Anh_P concentration

(d) $Disacch_{P,cyto}$ concentration

Figure 1: AmpC and modulator pools in the uncalibrated model

ative changes confined to the sixth decimal place or beyond, even after cefoxitin addition.

The activator muropeptides, $\mathrm{Anh_P}$ and $\mathrm{Disacch_{P,cyto}}$, behave differently: under cefoxitin exposure, both increase substantially, by approximately **66-fold** and **73-fold** respectively, over the simulated period. This response is masked in the figure only because both pools remain confined to sub-micromolar concentrations throughout, rendering a genuine two-order-of-magnitude change visually indistinguishable from noise at the plotted scale.

The failure of AmpC to show corresponding induction therefore does not reflect an absence of pathway activity, but rather a magnitude mismatch between the responding activator signal and the static repressor signal governing the AmpR switch. UDP-NAM-pentapeptide, the repressor ligand, remains fixed at approximately 1.25 mM throughout, roughly sixfold above its independently measured physiological target of $\sim$0.2 mM (Mengin-Lecreulx et al., 1982), because the uncalibrated Und-P recycling parameters (govern-

ing BacA, DedA, and de novo Und-P synthesis) impose a flux bottleneck that the original, literature-sourced rate constants do not resolve. Even a 66-fold increase in Anh_P leaves its absolute concentration several orders of magnitude below its dissociation constant with AmpR, so its contribution to the transcriptional switch remains negligible relative to the artificially elevated and unchanging repressor term. The model's peptidoglycan-recycling and drug-response machinery is therefore functioning qualitatively as intended, but the uncalibrated parameter set produces a repressor pool far outside its physiological range, which is sufficient on its own to suppress any downstream induction signal irrespective of genotype or drug exposure. This motivates the calibration procedure described in the following section.

8.2 Simulation Results of Manually Calibrated Model

8.2.1 Parameters Calibrated

Table 1: Parameter changes made during model calibration

Category	**Parameter**	**Original**	**Calibrated**
Und-P recycling	v_{de_novo}	2.40×10^{-4}	5.00×10^{-3}
	$V_{max,BacA}$	1.20×10^{-5}	2×10^{-2}
	$V_{max,DedA}$	6.75×10^{-4}	2×10^{-2}
	$V_{max,MraY}$	5.97×10^{-4}	7.043×10^{-4}
	$V_{max,MurG}$	7.01×10^{-5}	3.50×10^{-3}
Periplasmic synthesis / crosslinking	$k_{cat,PBP5}$	3.47	9.29×10^{-4}
	$k_{cat,PBP6}$	3.47	9.29×10^{-4}
	$V_{max,RodA}$	1.11×10^{-2}	8.80×10^{-4}
	$k_{cat,PBP1a,GT}$	100	7.93
Lytic transglycosylase	$V_{max,Slt}$	2.50×10^{-3}	4.8939×10^{-3}

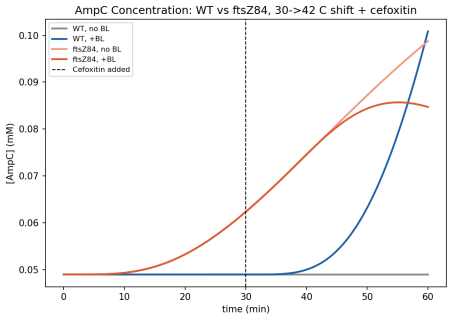
- v_{de_novo} — increased after tracing the Und-P recycling chain back to its source: total Und-PP production capped downstream flux regardless of any downstream enzyme's own capacity.
- $V_{max,BacA}$ — increased to relieve substrate starvation identified as the first bottleneck in Und-PP-to-Und-P conversion.
- $V_{max,DedA}$ — increased because, once BacA was unblocked, periplasm-to-cytoplasm Und-P transport became the next limiting step, causing

Und-P to accumulate in the periplasm without reaching the cytoplasmic pool.

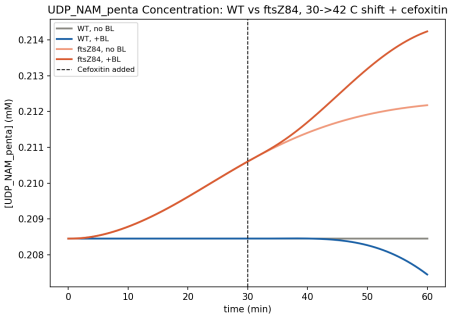
- $V_{max,MraY}$ — modestly increased once Und-P availability was restored, as MraY's own capacity had become mildly limiting relative to upstream UDP-NAM-pentapeptide production.
- $V_{max,MurG}$ — increased after diagnosis showed this step operating at ∼97% saturation, capping the entire downstream Lipid II supply chain independently of every other enzyme's capacity.
- $k_{cat,PBP5}$, $k_{cat,PBP6}$ — reduced to correct a combined carboxypeptidase capacity that vastly exceeded polymerization flux; calibrated to reproduce the independently measured 44–60% crosslinking fraction of the sacculus.
- $V_{max,RodA}$, $k_{cat,PBP1a,GT}$ — reduced jointly so that the FtsZ-dependent divisome enzymes (FtsW, PBP1b) represent a genuinely non-negligible share of total polymerization capacity, rather than having their loss fully compensated by RodA/PBP1a; calibrated to reproduce the ∼30% reduction in murein synthesis reported upon division block.
- $V_{max,Slt}$ — increased to raise activator muropeptide generation, set to the value maximizing wild-type fold-induction within this parameter's achievable range.

8.2.2 Results after Calibration

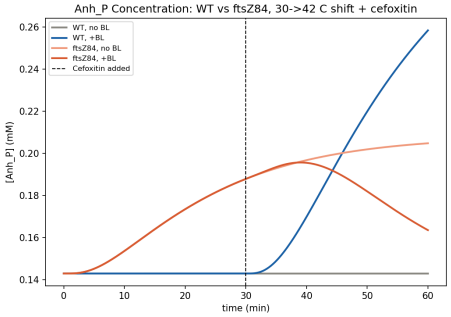
In Figure 2a, condition 2 (WT, +BL) shows AmpC rising from a shared baseline of 0.049 to 0.101 by one hour post-cefoxitin, a **2.06-fold induction**, which is a genuine, drug-dependent response, though still well short of the **6.2-fold target** reported by Uehara and Park. Comparing conditions 4 and 3 within the *ftsZ84* background, however, reveals a result distinct from the expected blunted-induction phenotype: AmpC decreases from 0.099 to 0.085 following cefoxitin addition, a fold-change of 0.86. Rather than reproducing a smaller but still positive induction (the ∼1.4-fold response reported in the literature), the calibrated model shows drug exposure actively suppressing AmpC relative to the *ftsZ84* no-drug baseline. **Although condition 4 does show lower AmpC than condition 2, consistent in direction with the hypothesis and experimental findings of Uehara and Park, this comparison is not itself the validating metric, the correct within-genotype fold-change ($\frac{AmpC_4}{AmpC_3} = 0.86$) instead indicates**



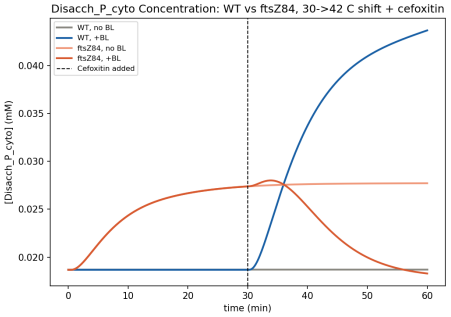
(a) AmpC concentration



(b) UMP concentration



(c) Anh_P concentration



(d) $Disacch_{P,cyto}$ concentration

Figure 2: AmpC and modulator pools in the manually calibrated model

that cefoxitin suppresses AmpC below the ftsZ84 baseline rather than producing the expected, smaller positive induction. More fundamentally, **the spurious rise in AmpC observed in conditions 3 and 4 in the absence of cefoxitin remains mechanistically unexplained**.

This indicates that the model's FtsZ-dependent gating is inadvertently altering peptidoglycan flux in a manner that elevates the activator signal independent of drug status. Resolving this will require identifying which FtsZ-gated flux term is responsible for the disproportionate activator response in the undrugged *ftsZ84* condition.

8.2.3 AmpC Fold-Induction: Model versus Literature

The following can be interpreted from Figure 3:

- **WT induction:** The model reaches a fold-change of **2.06**, against a

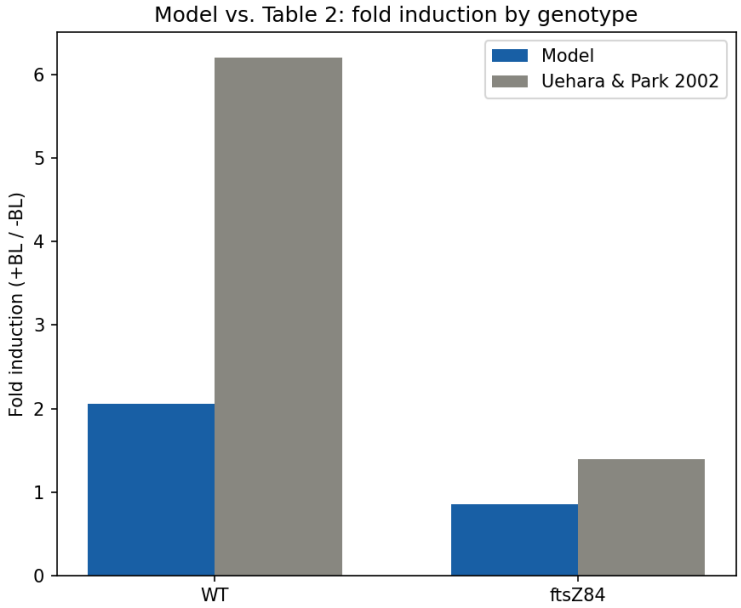


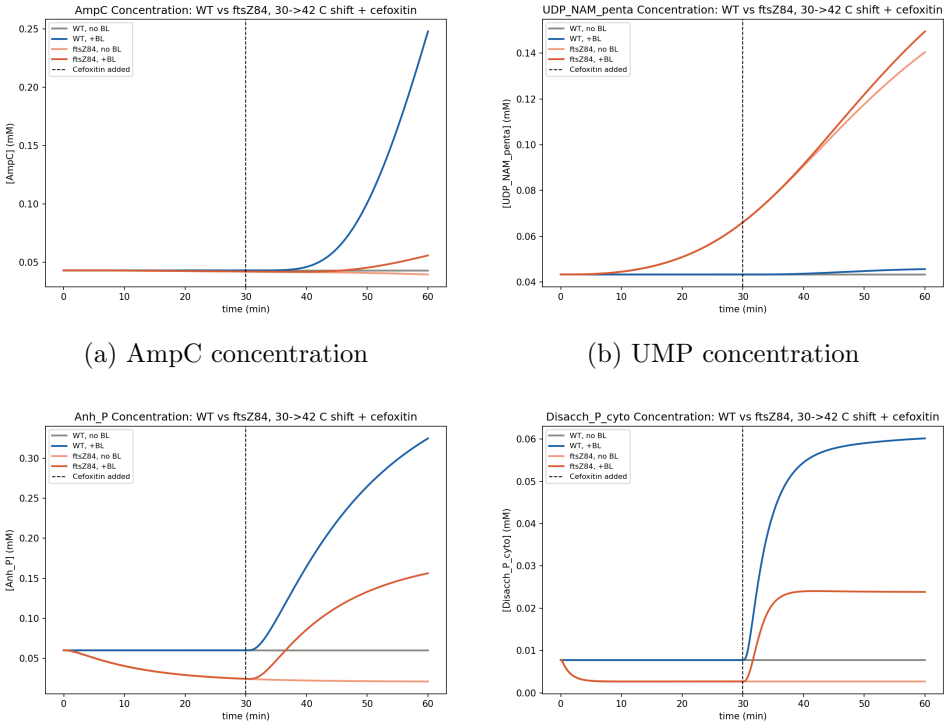
Figure 3: Fold Change Targets (Manually Calibrated Model)

target of **6.2**. Real, drug-dependent induction is occurring, but the magnitude is well short, at roughly a third of the target value.

- **ftsZ84 induction:** This is the more serious gap. The target is **1.4**, a smaller but still positive induction. The model gives **0.86**, meaning AmpC actually decreases upon cefoxitin exposure in this background, rather than increasing. This is a difference in direction, not just degree, since scaling a number up cannot turn a decrease into an increase.

8.3 Simulation Results of Automated Calibrated Model

These results align well with the expected outcomes. In Figure 4a, AmpC rises sharply and significantly from baseline in condition 2 (WT, +BL), consistent with the corresponding rise in activator concentrations shown in Figures 4c and 4d over the same timeframe and condition following cefoxitin addition. AmpC decreases by a significant margin in condition 4 (*ftsZ84*, +BL) relative to condition 2, consistent with the direction predicted by our hypothesis. This provides a visual demonstration of FtsZ inactivation affecting AmpC induction.



(a) AmpC concentration

(b) UMP concentration

(c) Anh_P concentration

(d) $Disacch_{P,cyto}$ concentration

Figure 4: AmpC and modulator pools in the automated calibrated model

The only quirk that remains unresolved is the repressor's behaviour running counter to expectations. The literature indicates that the repressor decreases in concentration under cefoxitin exposure (conditions 2 and 4) relative to the corresponding non-induced conditions (conditions 1 and 3), but Figure 4b shows the opposite trend: a small increase in repressor concentration in the drug-exposed conditions relative to their respective baselines. This points to a genuine gap in the current mechanistic understanding of AmpC induction and peptidoglycan biosynthesis. Since the model incorporates every known reaction across the pathways involved, this discrepancy may indicate a missing pathway or mechanism by which the repressor decreases under β-lactam stress.

Figure 5 shows the extent to which model predictions meet the literature targets for AmpC induction. The model predicts an AmpC fold-change of 5.7 against the literature-established target of 6.2, a marked improvement over the manually calibrated model's output (Figure 3). Similarly, the

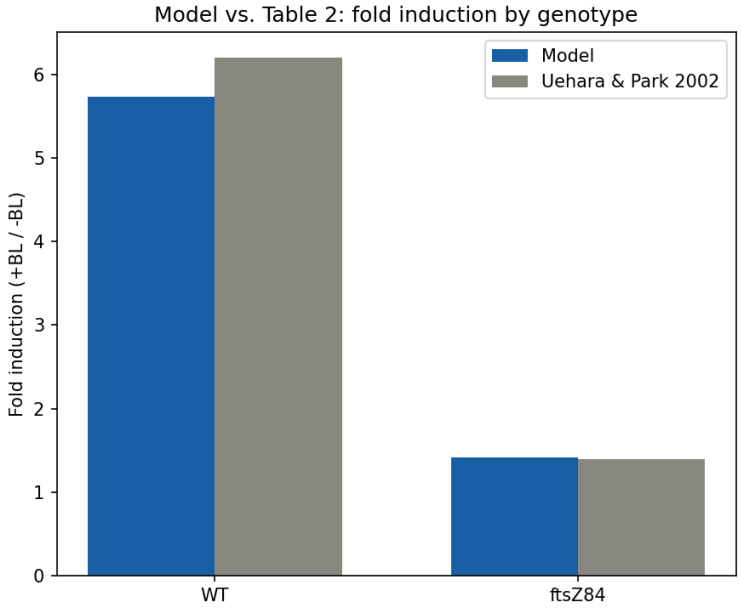
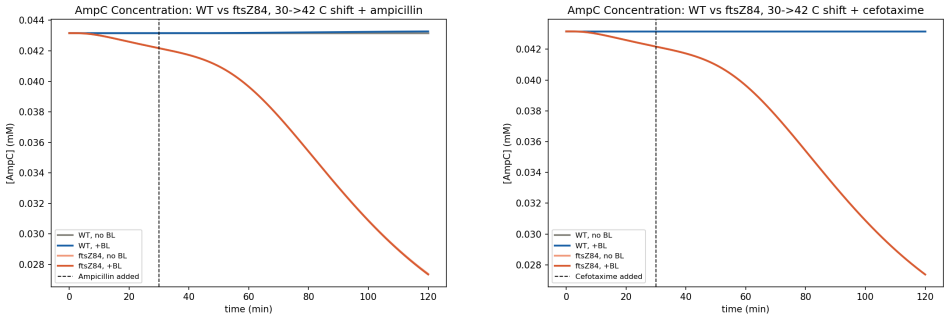


Figure 5: Fold Change Targets (Automated Calibrated Model)

model predicts an AmpC fold-change of 1.41 in the ftsZ84 genotype, closely matching the literature target of 1.40. These results indicate that the model reproduces the AmpC induction targets well, notwithstanding the repressor discrepancy noted above

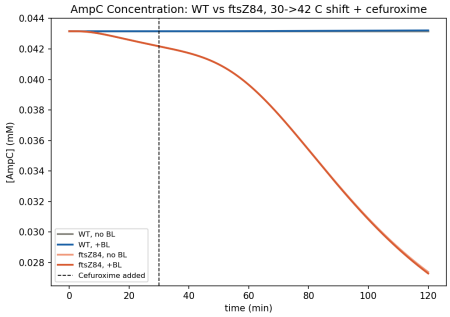
8.4 Simulation Results of the Model Tested on Different Drugs

Across all five compounds (ampicillin, cefotaxime, cefuroxime, cephalexin, oxacillin), the predicted AmpC trajectories are visually indistinguishable from one another (Figure 6). Wild-type AmpC remains flat irrespective of drug, and the ftsZ84 background shows the same 37% decline over 120 min regardless of which compound is added. This is notable given that the five compounds span more than two orders of magnitude in k_{acyl} at their respective dominant PBP targets and differ in which PBP each principally acylates (PBP3 for cefotaxime/cefuroxime, PBP4 for cephalexin, PBP3/PBP4 jointly for oxacillin, a broader multi-PBP profile for ampicillin). The model output shows no detectable sensitivity to any of this. The ftsZ84 mutation term alone appears to be driving the entire AmpC decline, with the

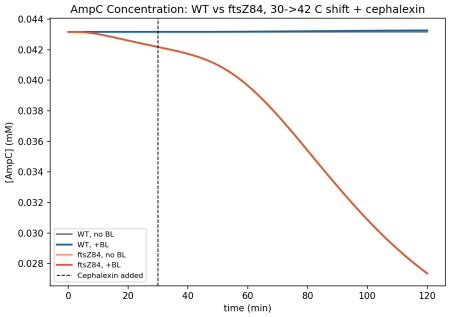


(a) Ampicillin

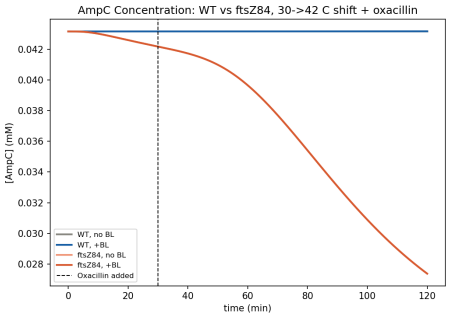
(b) Cefotaxime



(c) Cefuroxime



(d) Cephalexin



(e) Oxacillin

Figure 6: AmpC concentrations, as predicted by the model, under different drugs

drug-specific acylation kinetics contributing nothing distinguishable. This indicates a failure to translate PBP-level kinetic differences into distinct

AmpC responses, and should be treated as a model limitation (parameters not propagating into the relevant rate terms, or their effect being swamped by the genotype term) rather than a genuine finding that drug identity is irrelevant to AmpC dynamics.

A possible reason for this discrepancy could be that the model relies too heavily on PBP5/6 flux for AmpC induction. *E. coli* expresses PBP5/6 far more abundantly than PBP4, so carboxypeptidase activity there dominates the induction signal. Cefoxitin acylates PBP5/6 efficiently (k_{acyl} = 2.35 $\mathrm{mM^{-1}s^{-1}}$); all five drugs analysed here are essentially inert at this same target (k_{acyl} roughly four orders of magnitude weaker), despite several being potent at PBP3 or PBP4. This selective gap at PBP5/6, not general weakness against *E. coli* PBPs, is why the model predicts no induction for any of them.